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(54) DISPLAY DEVICE

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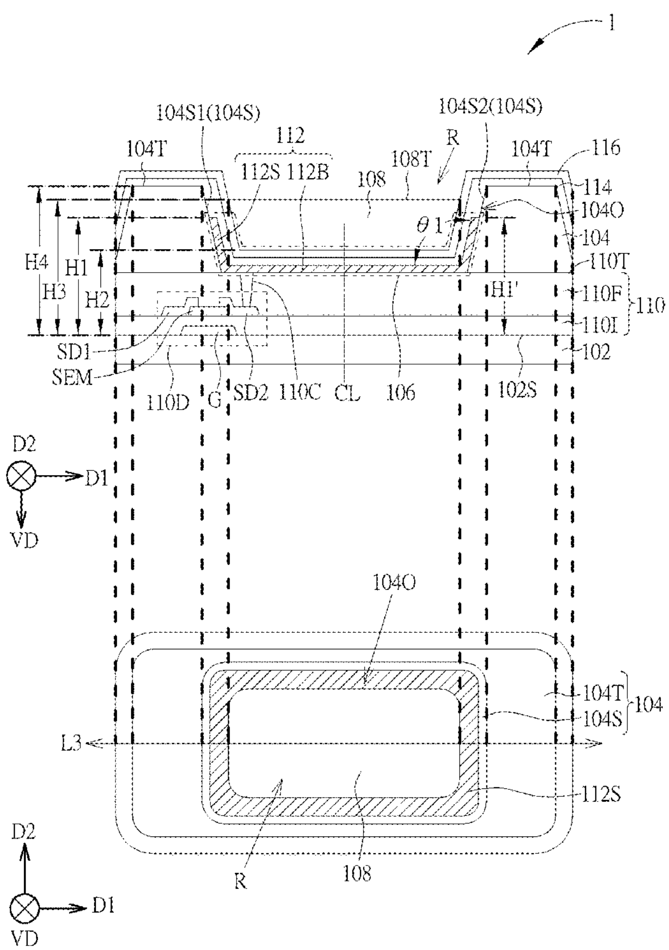
(57) **ABSTRACT**

A display device is provided and includes a substrate, an insulating layer disposed on the substrate and including at least one opening, a light blocking layer disposed on the insulating layer, a light emitting element disposed in the opening, a light conversion element on the light emitting element, and a color filter layer. In a cross-sectional view, the light blocking layer has a first maximum thickness, and the insulating layer has a second maximum thickness, wherein the first maximum thickness is greater than the second maximum thickness. In the cross-sectional view, the light conversion element has a third maximum thickness, and the color filter layer has a fourth maximum thickness, wherein the third maximum thickness is greater than the fourth maximum thickness.



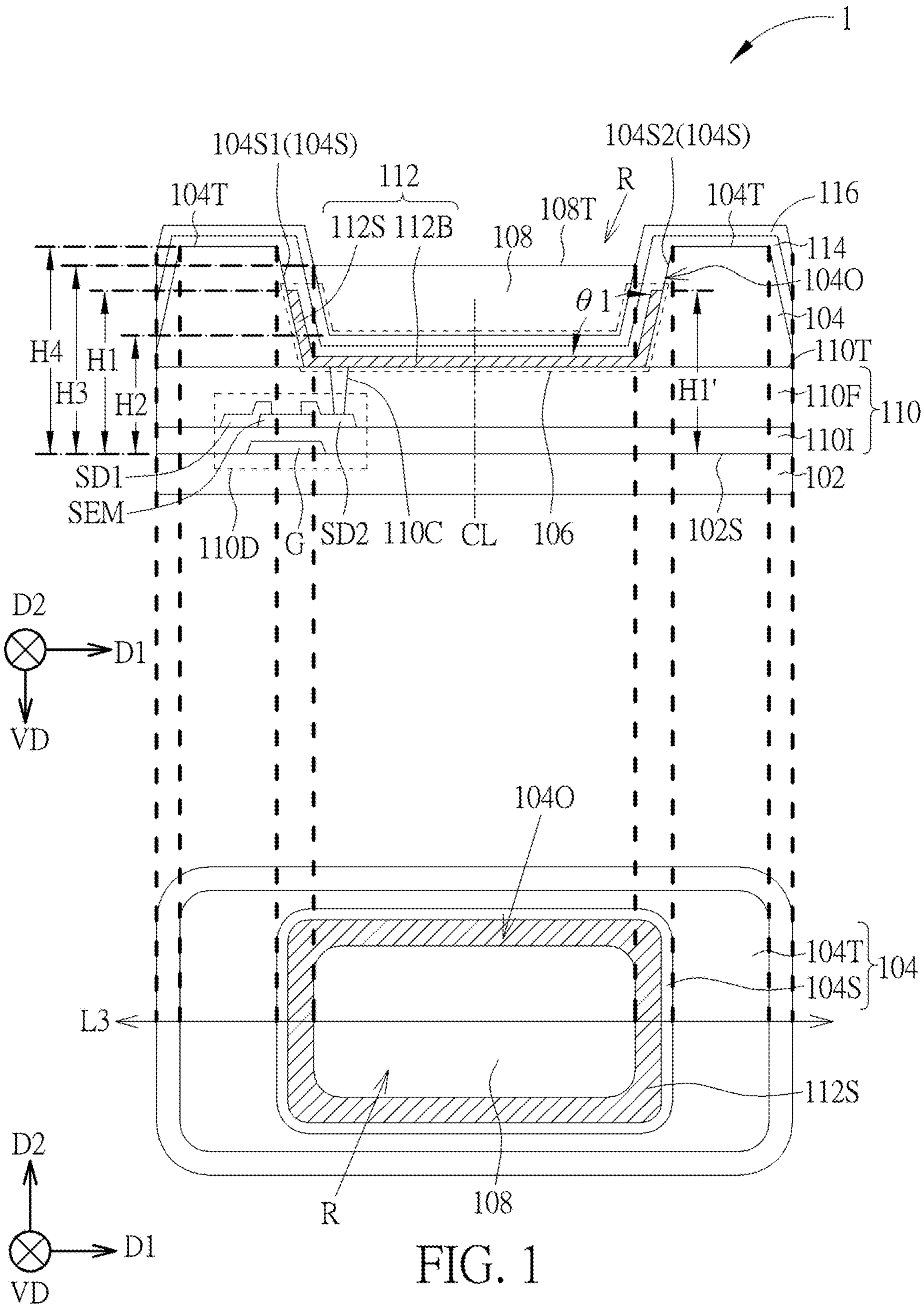
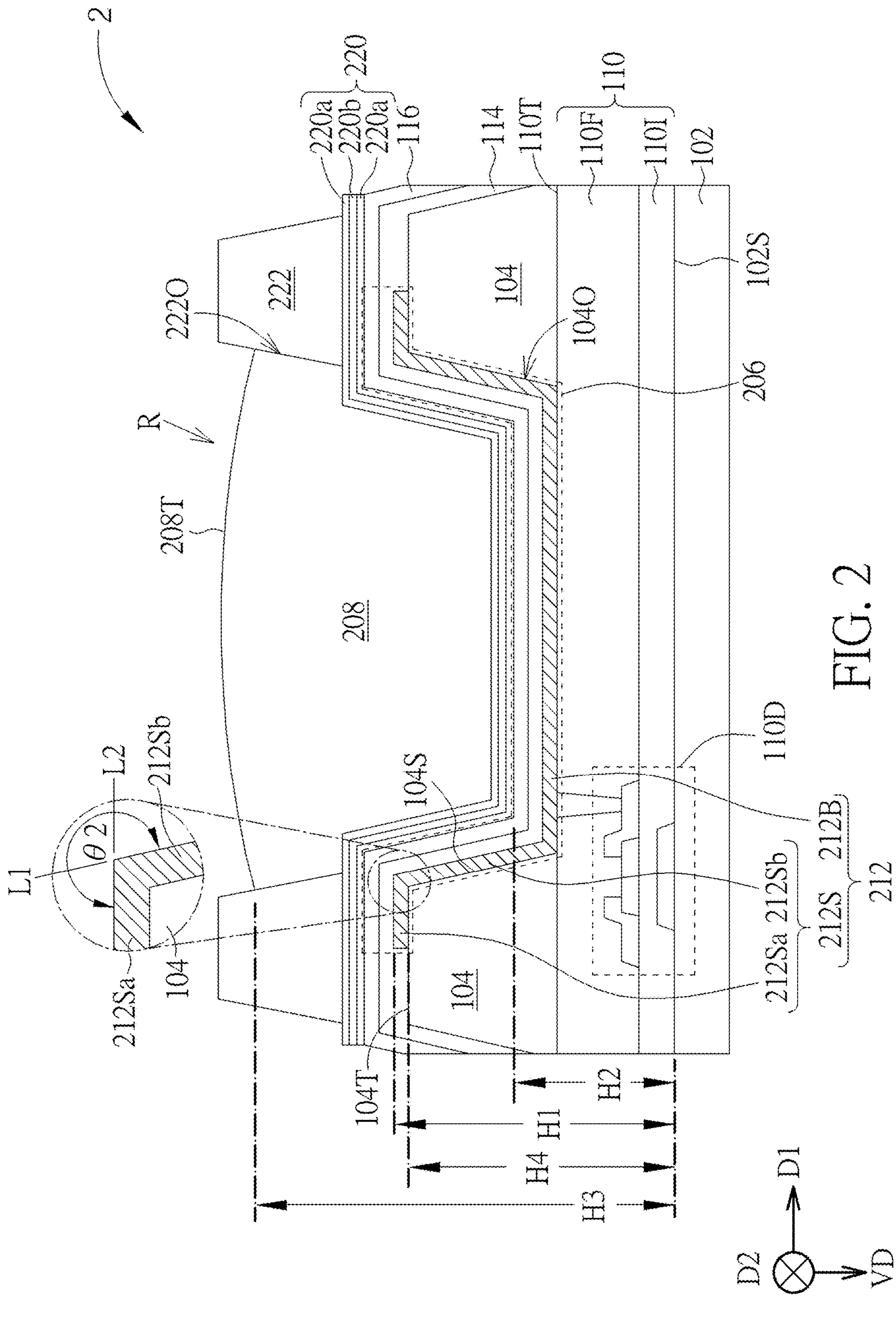

1
104S1(104S)
104S2(104S)
112
R
104T
116
112S 112B
108T
104T
108
114
104O
θ1
104
H4
H1
110T
H3
H2
H1'
110F
110
110I
SD1
102
SEM
102S
110D
G
SD2
110C
CL
106
D2
D1
VD
104O
104T
104
104S
L3
112S
R
108
D2
D1
VD

FIG. 1



2
220a
220b
220a
220
116
114
110T
110F
110
110I
102
102S
222
222O
104
104O
206
R
208T
208
110D
212B
212
212S
212Sa
212Sb
104S
L1
L2
θ2
212Sb
212Sa
104
104
104T
H1
H2
H3
H4
D1
D2
VD

FIG. 2

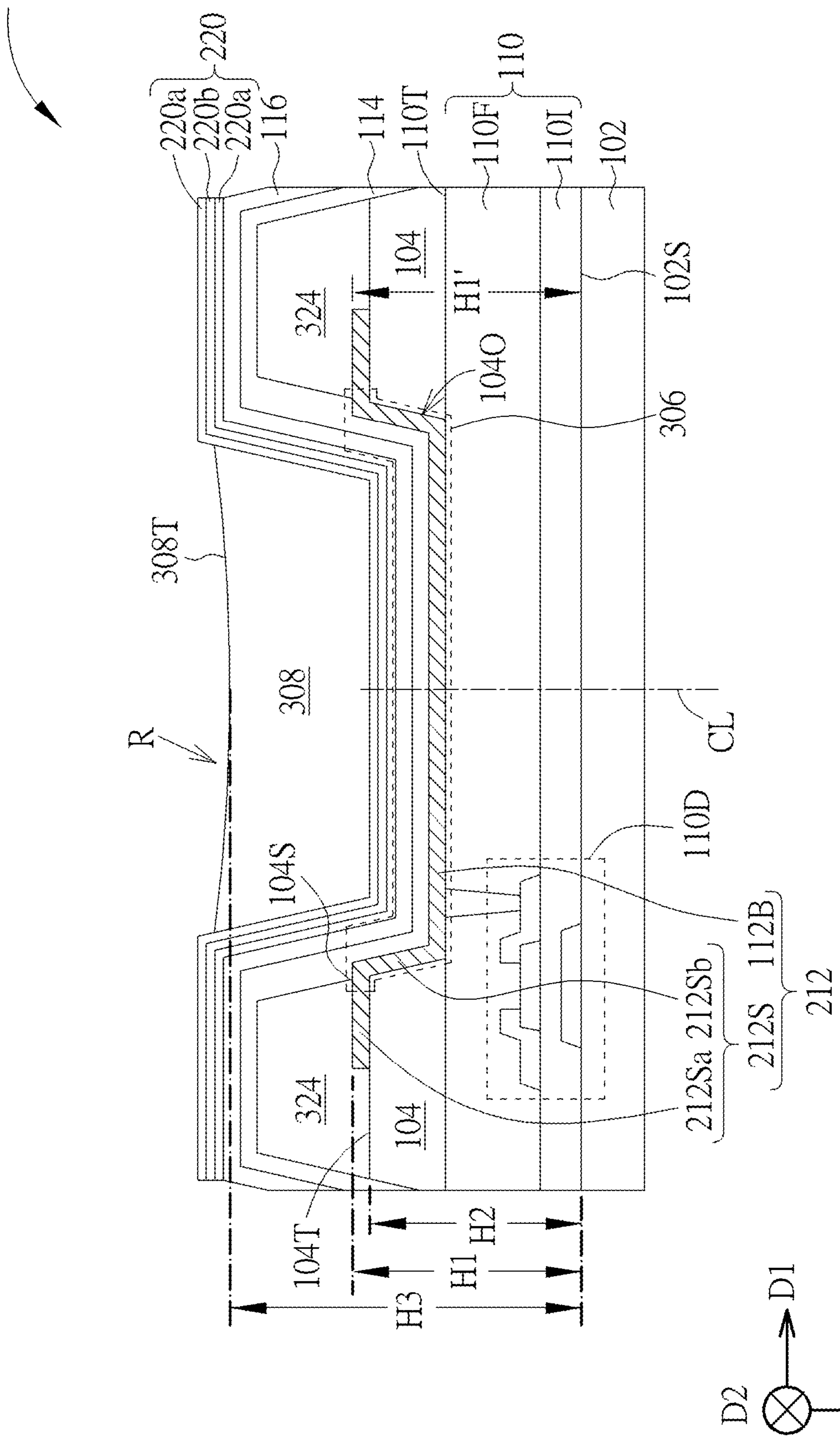

220
220a
220b
220a
116
114
110T
110F
110
110I
102
104
324
H1'
102S
104O
306
308T
308
R
CL
110D
104S
112B
212Sb
212
212Sa
212S
104T
H2
H1
H3
D1
D2
VD

FIG. 3

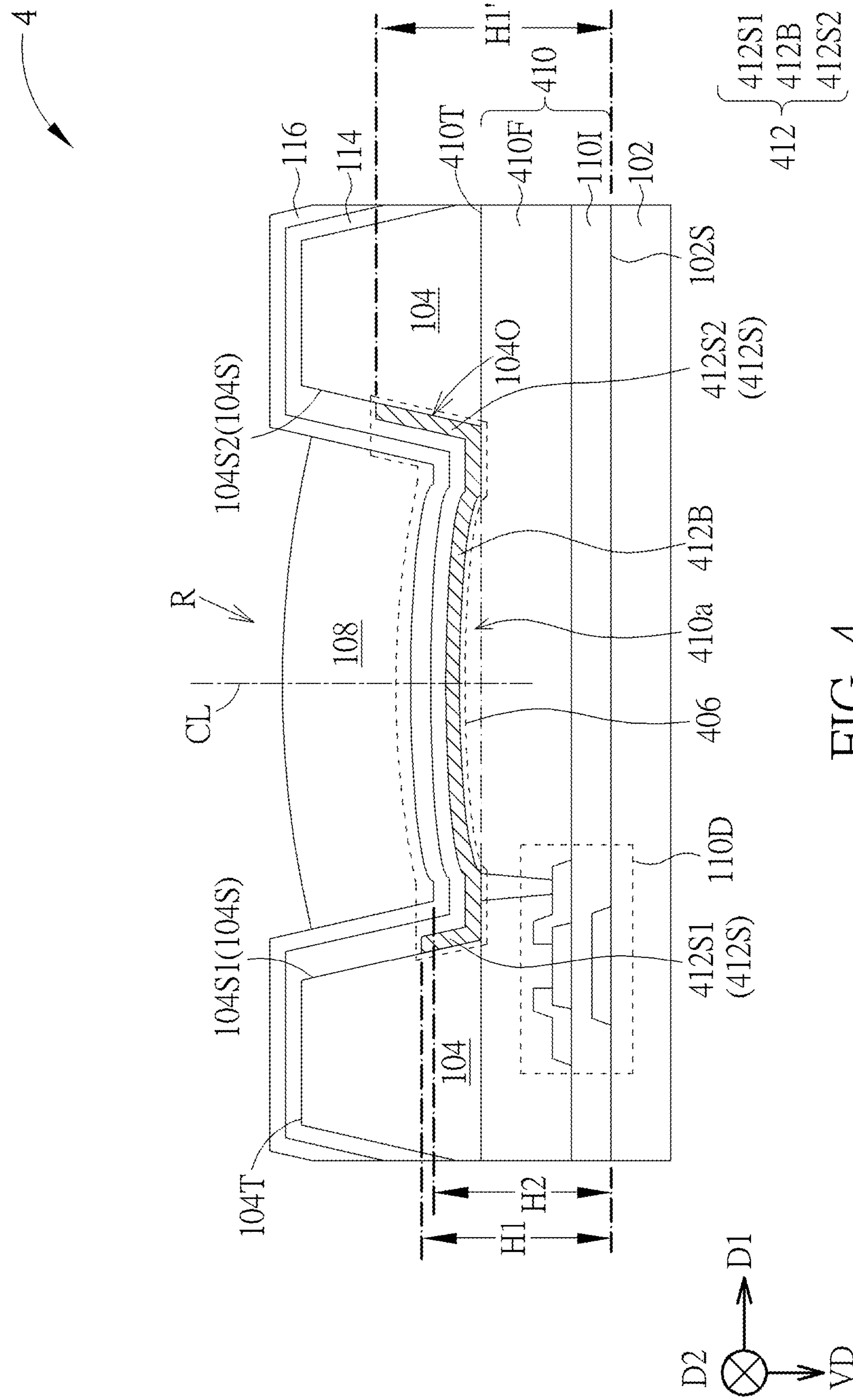

4
H1'
410
410T
410F
110I
102
102S
116
114
104
104O
412S2
(412S)
412S1
412B
412S2
412
104S2(104S)
412B
410a
R
108
CL
406
110D
104S1(104S)
412S1
(412S)
104
104T
H1
H2
D1
D2
VD

FIG. 4

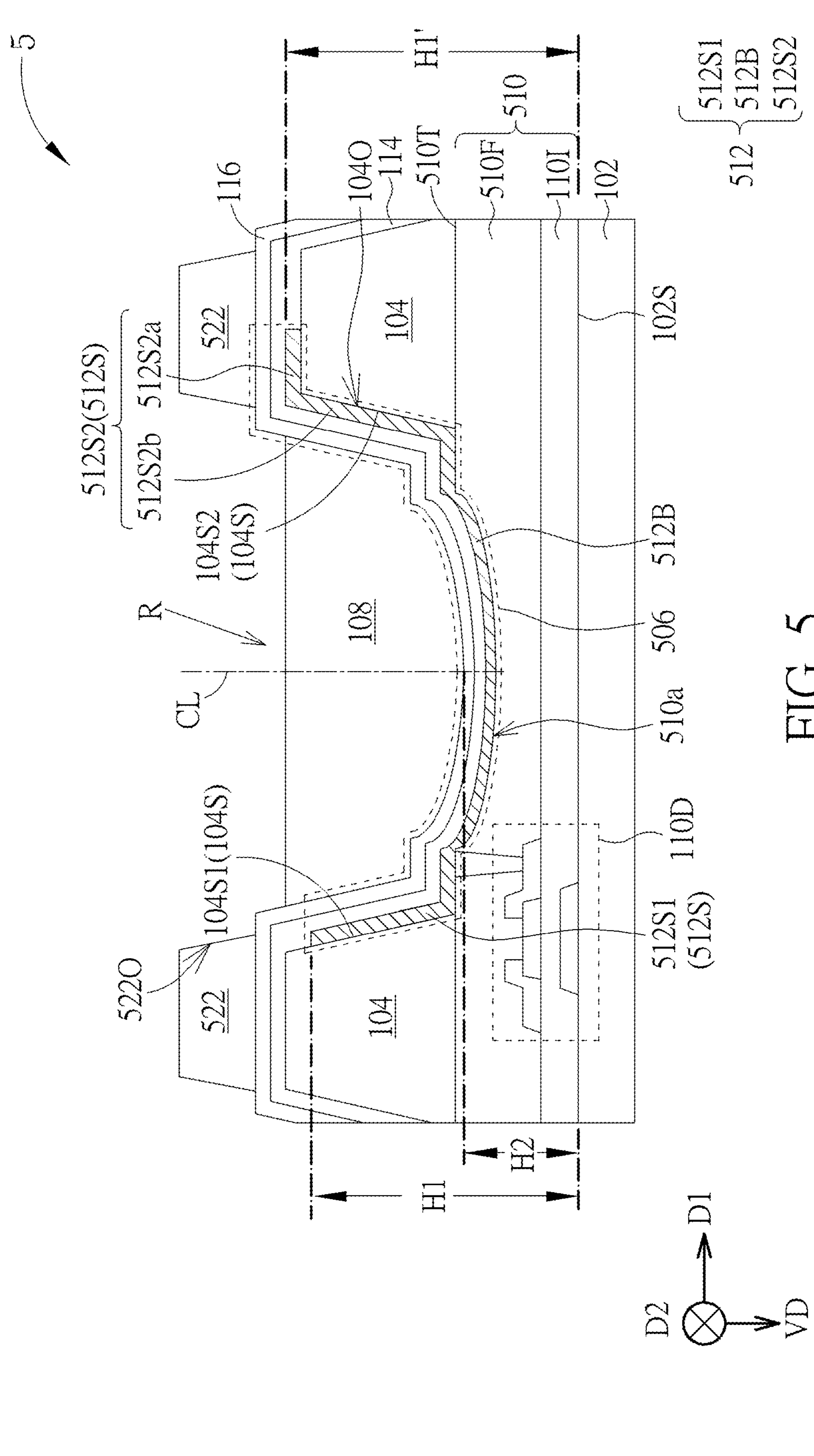

5
H1'
510
510F
110I
102
102S
510T
114
104O
116
104
522
512S2(512S)
512S2a
512S2b
104S2
(104S)
512B
108
R
506
CL
510a
110D
104S1(104S)
512S1
(512S)
522O
H1
H2
512
512S1
512B
512S2
D1
D2
VD

FIG. 5

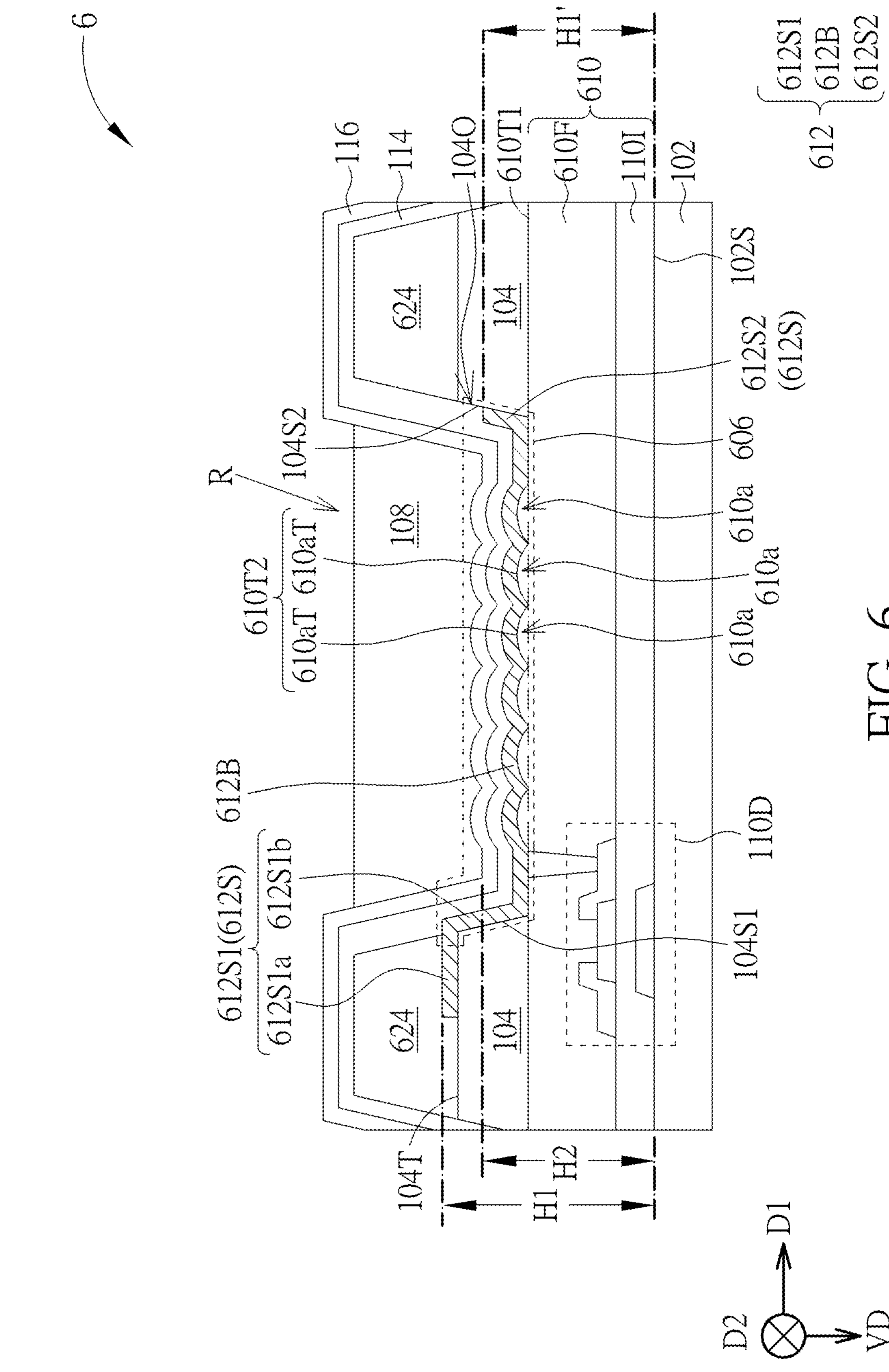

6
H1'
610
612S1
612B
612S2
612
116
114
104O
610T1
610F
110I
102
102S
624
104
612S2
(612S)
606
104S2
R
108
610a
610a
610a
610a
610T2
610aT 610aT
612B
110D
612S1(612S)
612S1a 612S1b
104S1
624
104
104T
H2
H1
D1
D2
VD

FIG. 6

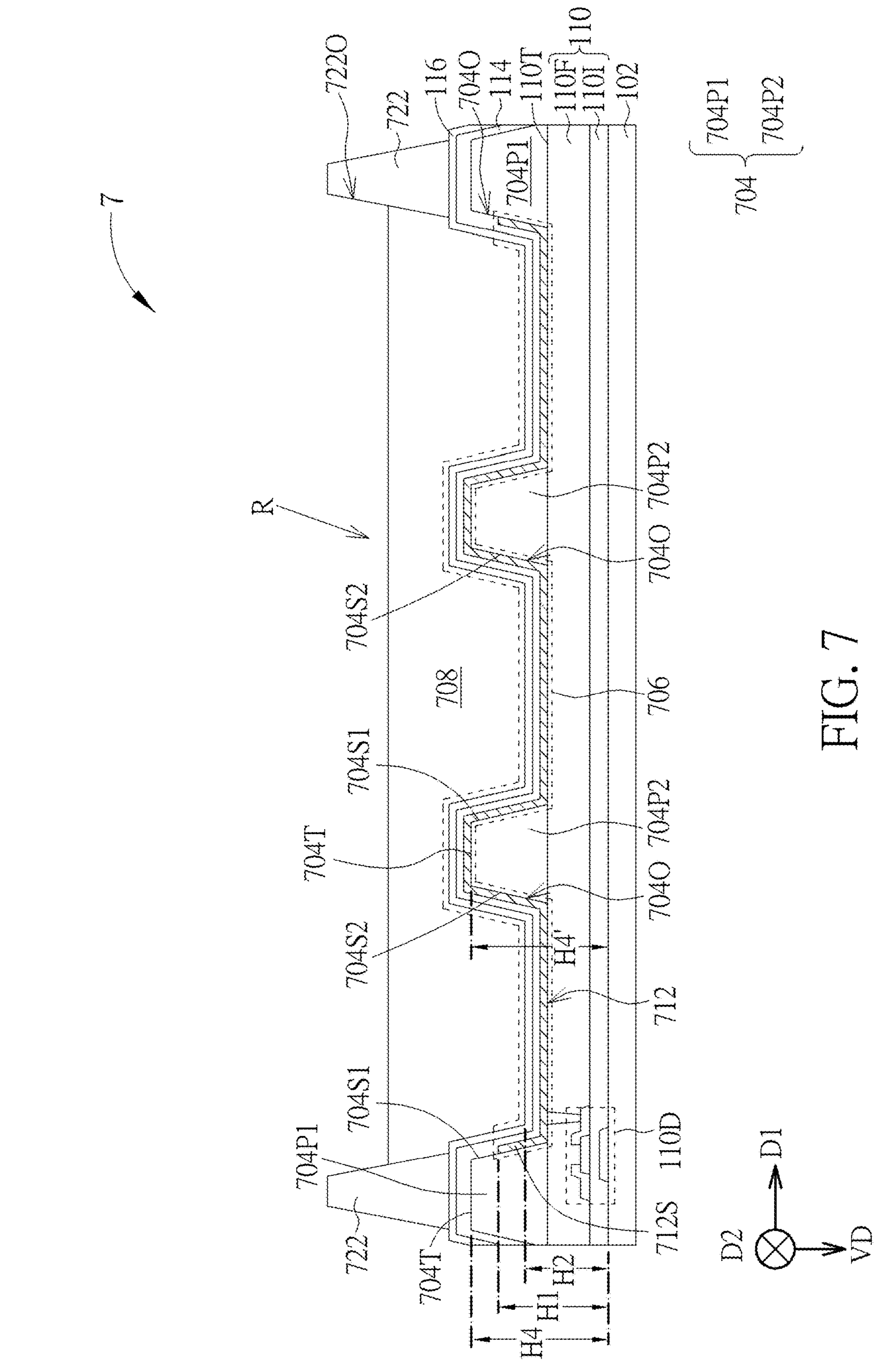

7
R
722O
722
116
704O
114
110T
110F
110I
110
102
704P1
704P2
704
704S2
708
704S1
704T
706
712
712S
110D
H4'
H4
H1
H2
D1
D2
VD

FIG. 7

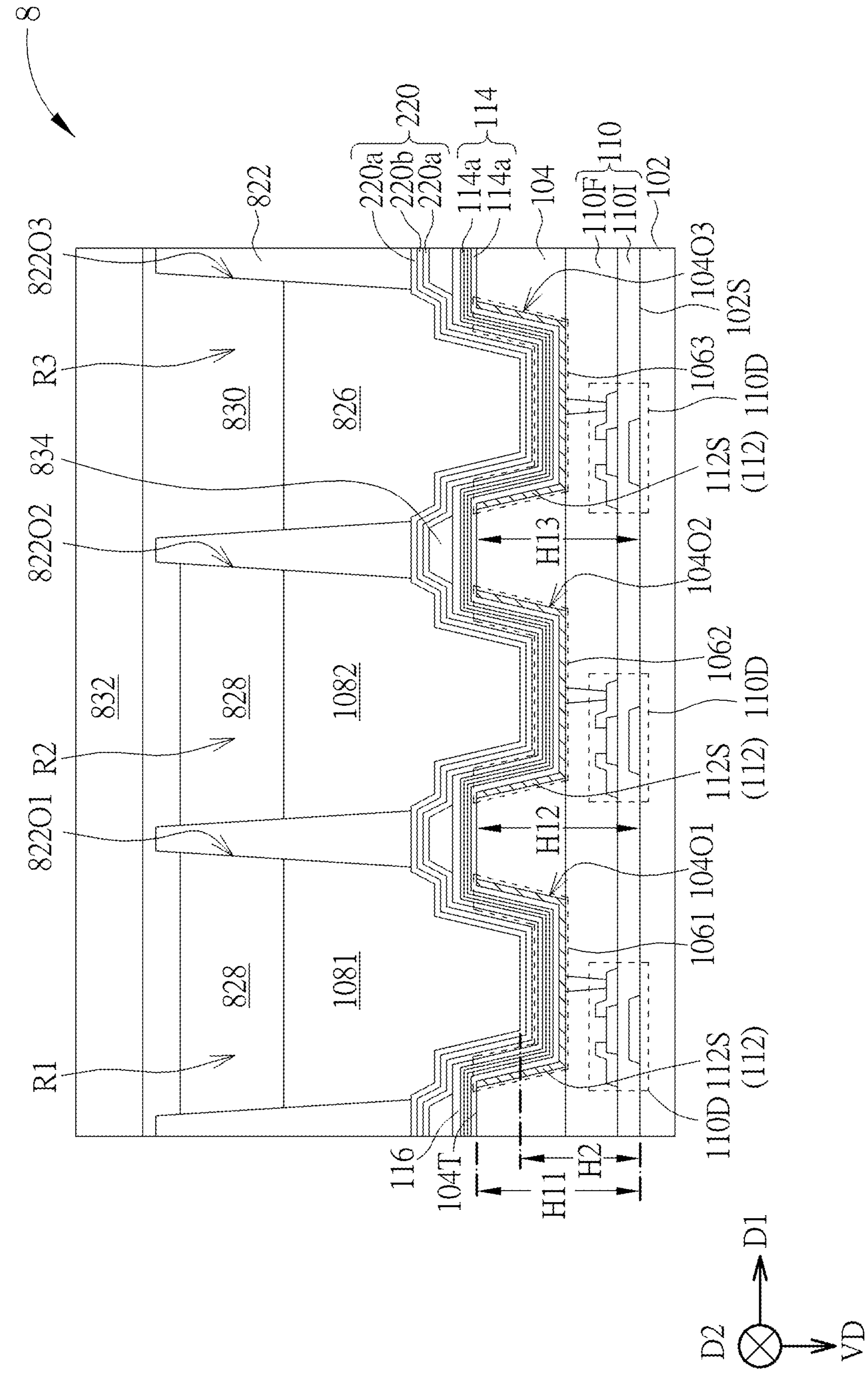
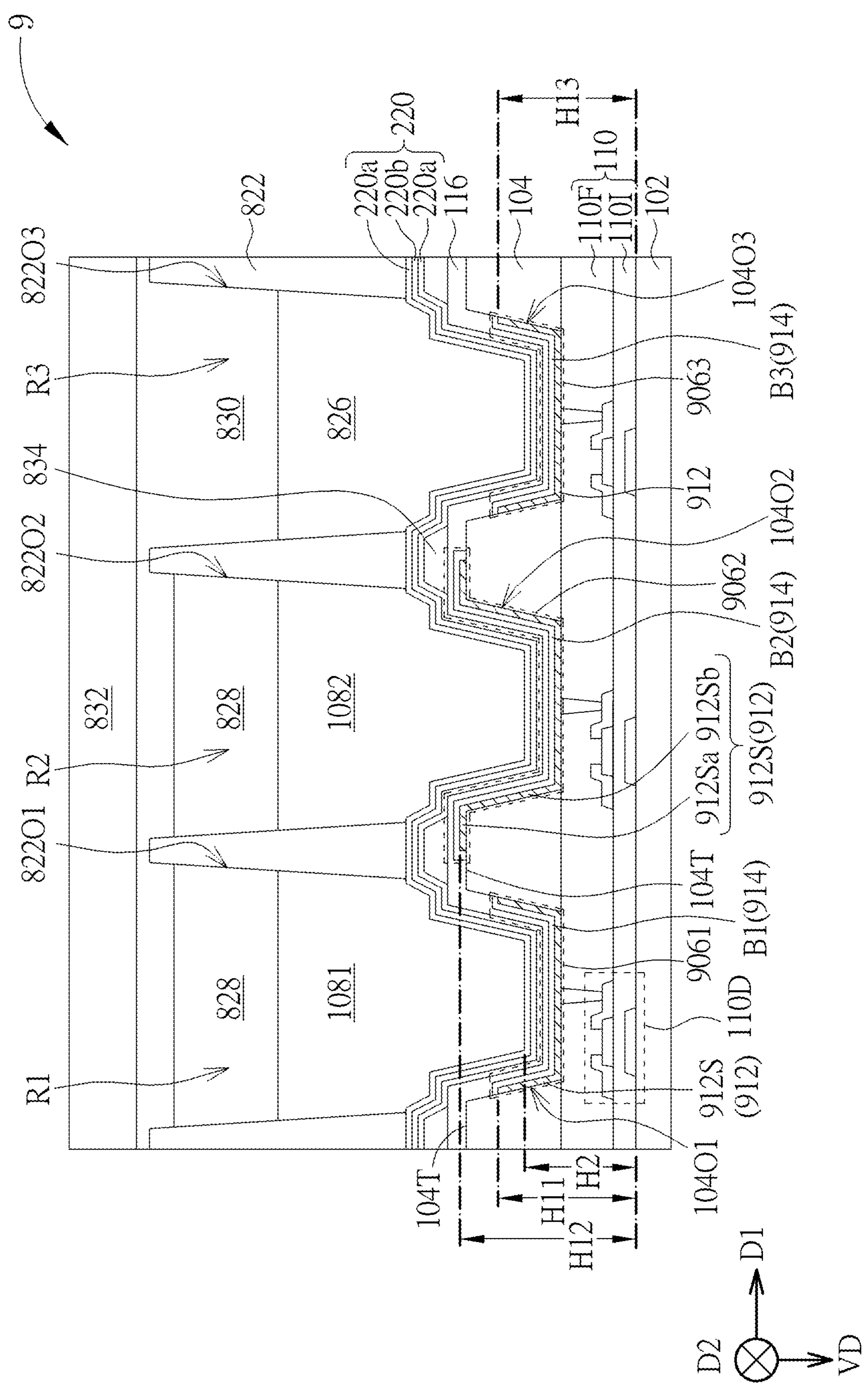

8
822O3
R3
834
822O2
822O1
R2
R1
822
830
826
832
828
1082
828
1081
220a
220b
220a
220
114a
114a
114
104
110F
110I
110
102
104O3
102S
1063
110D
112S
(112)
H13
104O2
1062
110D
112S
(112)
H12
104O1
1061
112S
(112)
110D
116
104T
H11
H2
D1
D2
VD

FIG. 8


9
822O3
822
220a
220b
220a
220
116
104
110F
110I
110
102
H13
104O3
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9063
912
104O2
9062
B2(914)
R3
830
826
834
822O2
832
828
1082
R2
822O1
912Sa 912Sb
912S(912)
104T
B1(914)
9061
110D
828
1081
R1
912S
(912)
104O1
104T
H11
H2
H12
D1
D2
VD

FIG. 9

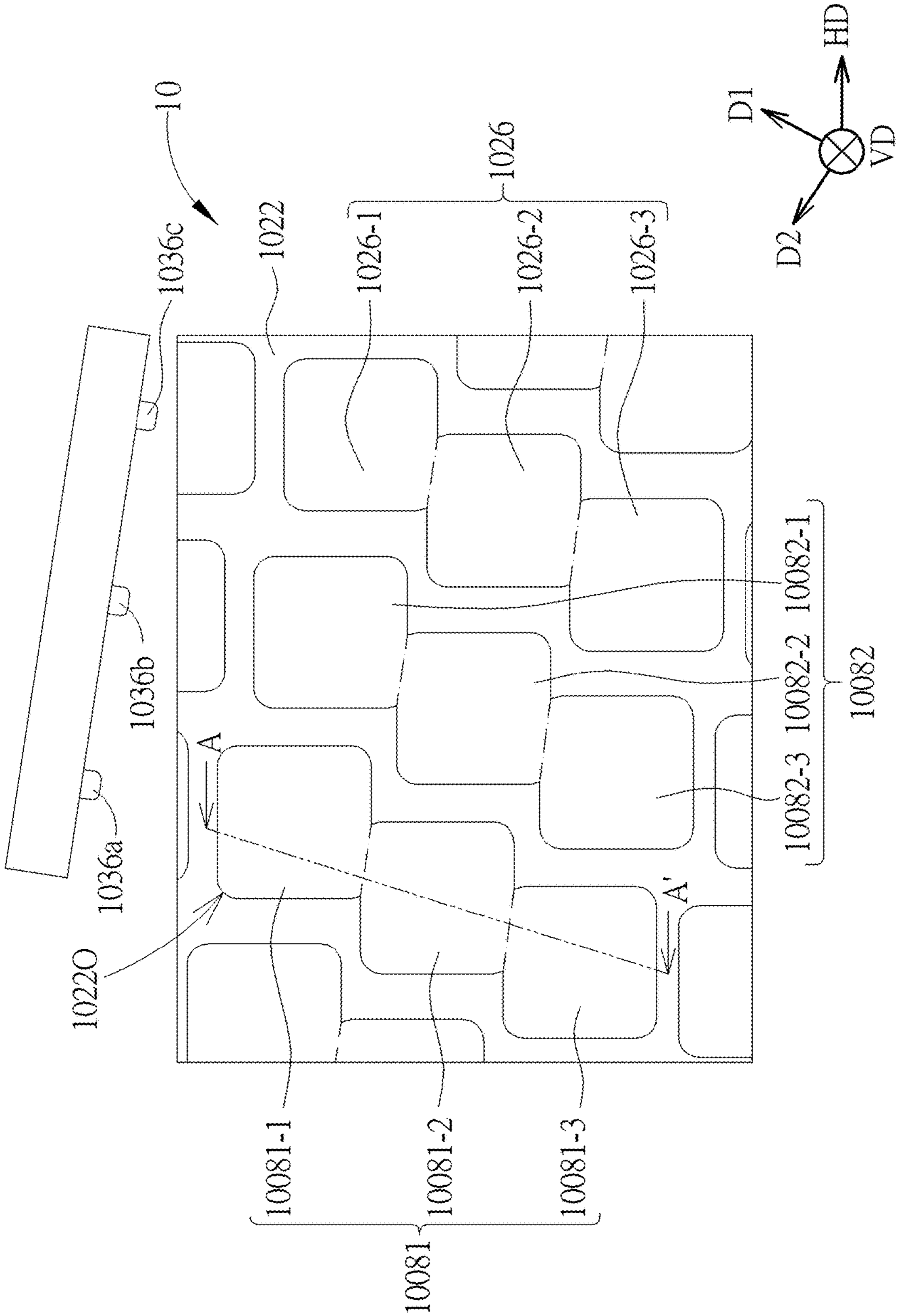
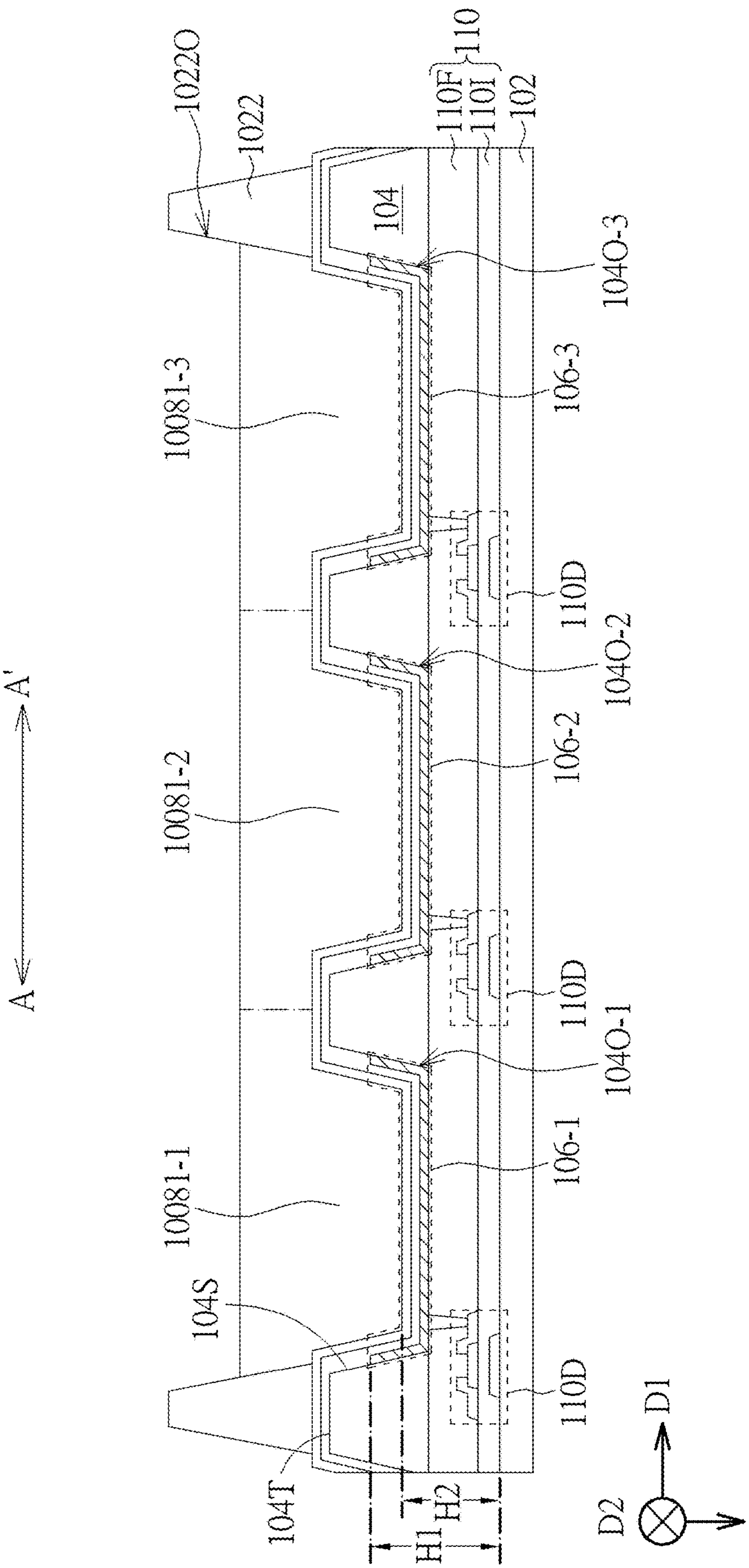

10
1036c
1036b
1036a
1022O
1022
1026
1026-1
1026-2
1026-3
10082
10082-1
10082-2
10082-3
10081
10081-1
10081-2
10081-3
A
A'
HD
D1
D2
VD

FIG. 10


1022O
1022
110F
110
110I
102
104
104O-3
10081-3
106-3
110D
104O-2
A'
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10081-2
106-2
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10081-1
106-1
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110D
D1
104T
H1
H2
D2
VD

FIG. 11

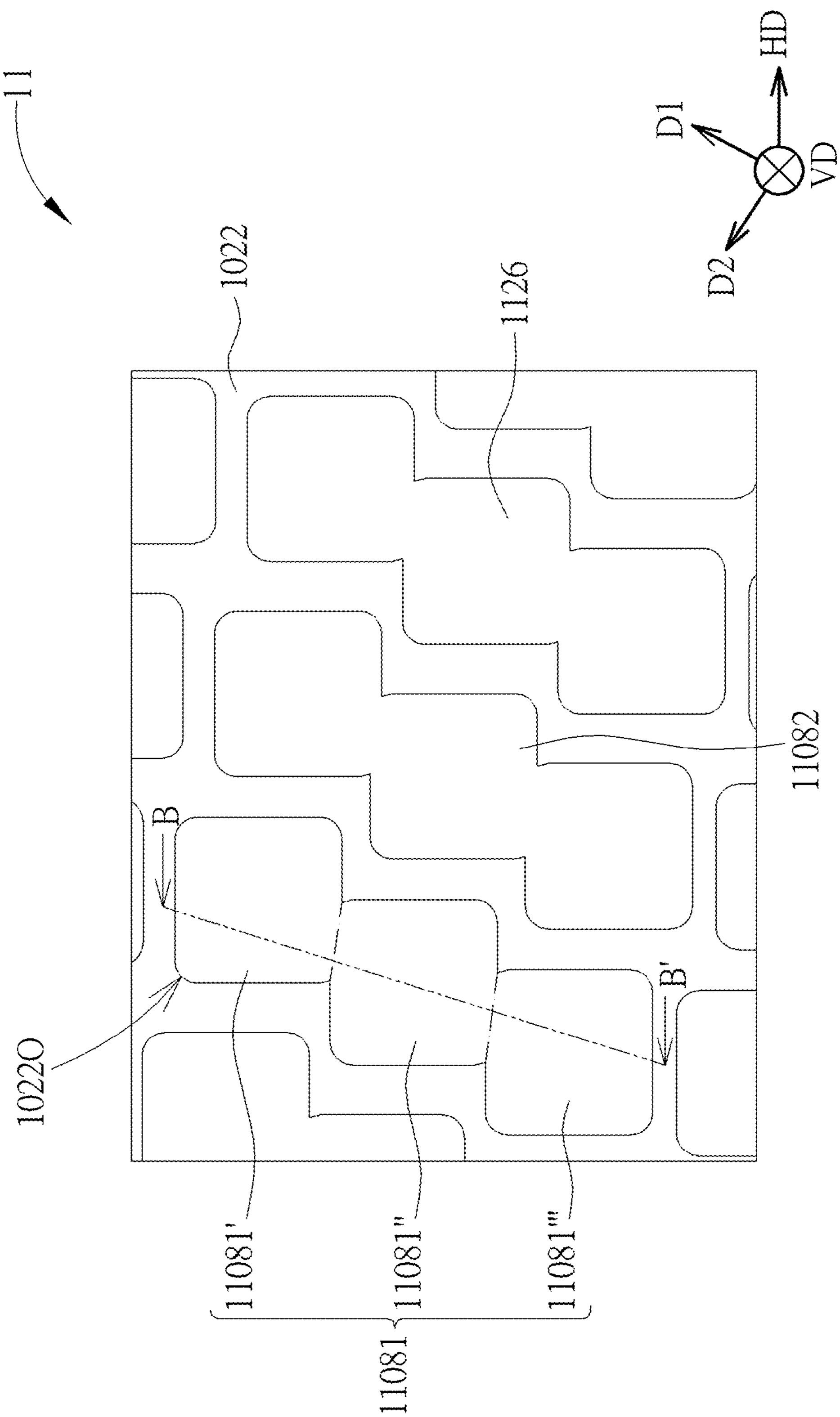
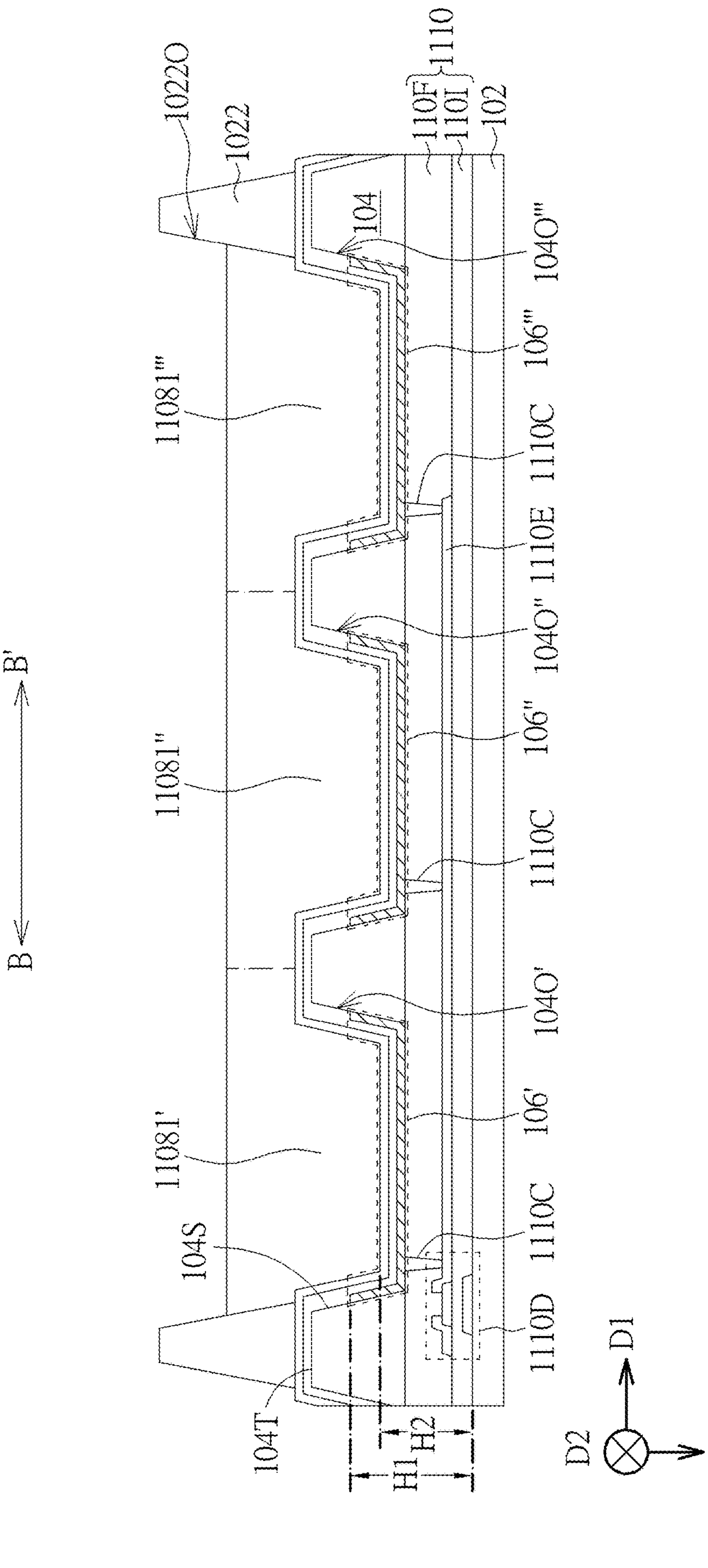

11
HD
D1
VD
D2
1022
1126
11082
B
B'
1022O
11081'
11081"
11081'"
11081

FIG. 12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1022
110F
110I
1110
102
104
104O'''
106'''
11081'''
1110C
1110E
104O''
106''
11081''
B
B'
1110C
104O'
106'
11081'
1110C
104S
1110D
104T
H1
H2
D1
D2
VD

FIG. 13

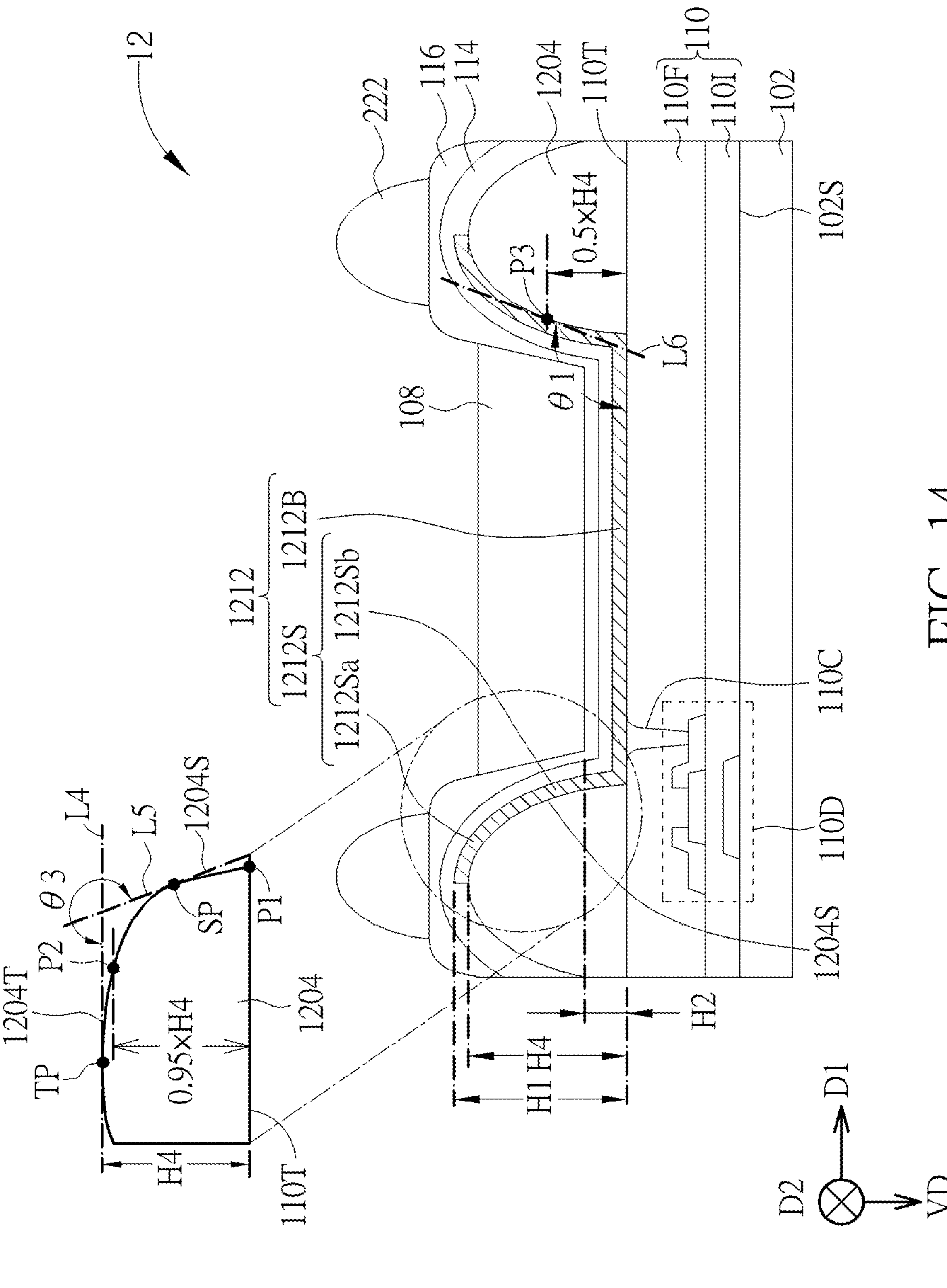
12
222
116
114
1204
110T
110F
110
110I
102
102S
0.5×H4
P3
L6
θ1
108
1212
1212S
1212B
1212Sa
1212Sb
110C
110D
1204S
H1
H4
H2
D1
D2
VD
TP
1204T
P2
θ3
L4
L5
1204S
SP
P1
0.95×H4
1204
H4
110T

FIG. 14

DISPLAY DEVICE

CROSS REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation application of U.S. application Ser. No. 18/393,704, filed on Dec. 22, 2023, which is a continuation application of U.S. application Ser. No. 17/857,052, filed on Jul. 4, 2022, which is a continuation application of U.S. application Ser. No. 16/988,723, filed on Aug. 10, 2020. The contents of these applications are incorporated herein by reference.

BACKGROUND OF THE DISCLOSURE

1. Field of the Disclosure

[0002] The present disclosure relates to a display device.

2. Description of the Prior Art

[0003] Quantum dots can convert excitation light emitted by a light emitting element into the converted light of desired color, and the converted light of different colors can be generated through adjusting sizes of the quantum dots. Therefore, in order to increase color saturation of display devices to improve image quality, manufacturers in the related fields propose a scheme of applying quantum dot materials to the display devices. However, the converted light generated by the quantum dots doesn't have directivity, so in the conventional display device, leakage or loss of converted light laterally emitted easily occurs, and the light emitting element as a light source also has some shortcomings. Accordingly, all quantum dots cannot be fully utilized, resulting in poor luminous efficiency of the display device.

SUMMARY OF THE DISCLOSURE

[0004] According to an embodiment of the present disclosure, a display device is disclosed and includes a substrate, an insulating layer, a light blocking layer, a light emitting element, a light conversion element, a color filter layer, and a protection layer. The insulating layer is disposed on the substrate and includes at least one opening. The light blocking layer is disposed on the insulating layer. At least a part of the light emitting element is disposed in the at least one opening. The light conversion element is disposed on the light emitting element. The color filter layer is disposed on the light conversion element, and the protection layer is disposed between the light conversion element and the light emitting element. In a cross-sectional view, the light blocking layer has a first maximum thickness, and the insulating layer has a second maximum thickness, wherein the first maximum thickness is greater than the second maximum thickness. In the cross-sectional view, the light conversion element has a third maximum thickness, and the color filter layer has a fourth maximum thickness, wherein the third maximum thickness is greater than the fourth maximum thickness.

[0005] These and other objectives of the present disclosure will no doubt become obvious to those of ordinary skill in the art after reading the following detailed description of the embodiment that is illustrated in the various figures and drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0006] FIG. **1** schematically illustrates a cross-sectional view and a corresponding top view of a display device according to a first embodiment of the present disclosure.

[0007] FIG. **2** schematically illustrates a cross-sectional view of a display device according to a second embodiment of the present disclosure.

[0008] FIG. **3** schematically illustrates a cross-sectional view of a display device according to a third embodiment of the present disclosure.

[0009] FIG. **4** schematically illustrates a cross-sectional view of a display device according to a fourth embodiment of the present disclosure.

[0010] FIG. **5** schematically illustrates a cross-sectional view of a display device according to a fifth embodiment of the present disclosure.

[0011] FIG. **6** schematically illustrates a cross-sectional view of a display device according to a sixth embodiment of the present disclosure.

[0012] FIG. **7** schematically illustrates a cross-sectional view of a display device according to a seventh embodiment of the present disclosure.

[0013] FIG. **8** schematically illustrates a cross-sectional view of a display device according to an eighth embodiment of the present disclosure.

[0014] FIG. **9** schematically illustrates a cross-sectional view of a display device according to a ninth embodiment of the present disclosure.

[0015] FIG. **10** schematically illustrates a top view of a display device according to a tenth embodiment of the present disclosure.

[0016] FIG. **11** schematically illustrates a cross-sectional view of FIG. **10** taken along a cross-sectional line A-A'.

[0017] FIG. **12** schematically illustrates a top view of a display device according to an eleventh embodiment of the present disclosure.

[0018] FIG. **13** schematically illustrates a cross-sectional view of FIG. **12** taken along a cross-sectional line B-B'.

[0019] FIG. **14** schematically illustrates a cross-sectional view of a display device according to a twelfth embodiment of the present disclosure.

DETAILED DESCRIPTION

[0020] The present disclosure may be understood by reference to the following detailed description, taken in conjunction with the drawings as described below, and for purposes of illustrative clarity and being easily understood by the readers, various drawings of this disclosure may be simplified, and the elements in various drawings may not be drawn to scale. In addition, the number and dimension of each element shown in drawings are just illustrative and are not intended to limit the scope of the present disclosure.

[0021] Certain terms are used throughout the description and following claims to refer to particular elements. As one skilled in the art will understand, electronic equipment manufacturers may refer to an element by different names. This document does not intend to distinguish between elements that differ in name but not function. In the following description and in the claims, the terms “comprise”, “include” and “have” are used in an open-ended fashion, and thus should be interpreted to mean “include, but not limited to . . . ”.

[0022] The direction terms used in the following embodiment such as up, down, left, right, in front of or behind are just the directions referring to the attached figures. Thus, the direction terms used in the present disclosure are for illustration, and are not intended to limit the scope of the present disclosure. It should be noted that the elements which are specifically described or labeled may exist in various forms for those skilled in the art. Besides, when a layer is referred to as being "on" another layer or substrate, it may be directly on the other layer or substrate, or may be on the other layer or substrate, or intervening layers may be included between other layers or substrates.

[0023] Besides, relative terms such as "lower" or "bottom", and "higher" or "top" may be used in embodiments to describe the relative relation of an element to another element labeled in figures. It should be understood that if the labeled device is flipped upside down, the element in the "lower" side may be the element in the "higher" side.

[0024] The ordinal numbers such as "first", "second", etc. are used in the specification and claims to modify the elements in the claims. It does not mean that the required element has any previous ordinal number, and it does not represent the order of a required element and another required element or the order in the manufacturing method. The ordinal number is just used to distinguish the required element with a certain name and another required element with the same certain name.

[0025] It should be noted that the technical features in different embodiments described in the following may be replaced, recombined, or mixed with one another to constitute another embodiment without departing from the spirit of the present disclosure.

[0026] The display device of the present disclosure may be an electronic device including a light emitting device, a sensing device or a tiled device, but not limited thereto. The display device may include a foldable display device or flexible display device. The tiled device may for example be a tiled display device, but not limited thereto. It is noted that the display device may be any combination of the above, but not limited thereto. The display device may be applied to electronic products or electronic devices that requires light source, light-emitting apparatus or display apparatus, for example but not limited to, television, tablet, notebook, mobile phone, camera, wearable device, electronic entertainment device, etc.

[0027] FIG. **1** schematically illustrates a cross-sectional view and a corresponding top view of a display device according to a first embodiment of the present disclosure, wherein an upper portion of FIG. **1** schematically depicts the cross-sectional view of the display device, and a lower portion of FIG. **1** schematically depicts the top view of the display device. For clearly illustrating the display device **1** of the present embodiment, the display device **1** in the upper portion of FIG. **1** omits layers disposed on the light conversion element, and the corresponding lower portion exemplifies a relative relationship between the first insulating layer **104** and the first electrode **112** of the light emitting element **106** of the display device **1** in a top view direction VD, but the present disclosure is not limited thereto. As shown in FIG. **1**, the display device **1** includes a substrate **102**, a first insulating layer **104**, a light emitting element **106**, and a light conversion element **108**. In the present disclosure, the light emitting element **106** is used for generating excitation light and emitting the excitation light to the light conversion element **108**, such that the light conversion element **108** may absorb the excitation light and convert the excitation light into converted light with a wavelength greater than a wavelength of the excitation light. The excitation light may for example be blue light or ultraviolet (UV), and the converted light may for example be red light, green light or blue light, but the present disclosure is not limited thereto. In the present embodiment, one light conversion element **108** may correspond to one light emitting element **106**, but not limited thereto. The term "corresponding to" used herein represents the one light conversion element **108** at least partially overlap the one light emitting element **106** in the top view direction VD. In some embodiments, one light conversion element **108** may correspond to a plurality of the light emitting elements **106**. In some embodiments, the display device **1** may include a plurality of light emitting elements **106** and a plurality of light conversion element **108** for generating converted light of different colors.

[0028] The substrate **102** may be used for carrying the first insulating layer **104**, the light emitting element **106**, the light conversion element **108** or other layers. The substrate **102** may be a rigid substrate or a flexible substrate. The material of the substrate **102** may for example include glass, quartz, sapphire, polyimide (PI), polycarbonate (PC), polyethylene terephthalate (PET), or combinations thereof, but not limited thereto. In the present embodiment, the display device **1** may further include a thin film transistor layer **110** disposed between the first insulating layer **104** and the substrate **102** and used for controlling a switch of the light emitting element **106** or a brightness of the generated excitation light. For example, the thin film transistor layer **110** may include at least one driving element **110**D and at least one switching element (not shown), and the light emitting element **106** may be controlled by the at least one driving element **110**D and the at least one switching element, but not limited thereto. The position of the driving element **110**D is not limited as shown in FIG. **1**, and in some embodiments, the driving element **110**D may for example be disposed under the light emitting element **106**. In some embodiments, the thin film transistor layer **110** may further include lines for controlling the display device **1**, such as scan lines, data lines or other driving circuits, but not limited thereto. The driving element **110**D shown in FIG. **1** takes bottom-gate type thin film transistor as an example, wherein the driving element **110**D may include a gate G, a semiconductor layer SEM, a source (drain) SD**1**, and a drain (source) SD**2**. The gate G is disposed between the semiconductor layer SEM and the substrate **102**, and the source (drain) SD**1** and the drain (source) SD**2** are disposed on two sides of the semiconductor layer SEM respectively. The thin film transistor layer **110** may further include an insulating layer **110**I regarded as a gate insulating layer of the driving element **110**D. The structure of transistor of the thin film transistor layer **110** is not limited to the mentioned above and may be a top-gate type transistor or be a dual-gate type transistor or other suitable transistors depending on the requirements. Or, the structure of the transistor of the thin film transistor layer **110** may for example be low-temperature polysilicon (LTPS) transistor, or low-temperature polycrystalline oxide (LTPO) complex transistor, but not limited thereto.

[0029] The first insulating layer **104** is disposed on the substrate **102** and includes at least one opening **104**O, a top surface **104**T, and a side surface **104**S, wherein the side surface **104**S surrounds the at least one opening **104**O, and the top surface **104**T is adjacent to the side surface **104**S. When the top surface **104**T is arc-shaped, method for distinguishing the top surface **104**T from the side surface **104**S may refer to the description corresponding to FIG. **14**. The first insulating layer **104** may for example be a pixel defining layer, and in a cross-sectional view, the first insulating layer **104** may exhibit at least one bump shape and be used for defining a pixel or sub-pixel of the display device **1**. For example, the opening **104**O of the first insulating layer **104** may correspond to one pixel or sub-pixel, such that light emitted from the opening **104**O in a direction opposite to the top view direction VD may be regarded as light of the pixel or sub-pixel, but not limited thereto. The first insulating layer **104** may include organic material, such as polyimide, but not limited thereto. In the present embodiment, the opening **104**O of the first insulating layer **104** and the thin film transistor layer **110** may form a recess R. For example, the thin film transistor layer **110** may include a planarization layer **110**F disposed on the driving element **110**D, and the first insulating layer **104** may be directly formed on the planarization layer **110**F, such that the first insulating layer **104** and the planarization layer **110**F may form the recess R. The planarization layer **110**F may be an insulating layer, and the material of the planarization layer **110**F may for example include organic material, but not limited thereto. Furthermore, the planarization layer **110**F may be used for electrically insulating the driving element **110**D from the light emitting element **106**, and the thin film transistor layer **110** may further include a connecting hole **110**C penetrating through the planarization layer **110**F or other insulating layers on the driving element **110**D, such that conductive material disposed in the connecting hole **110**C may electrically connect the driving element **110**D to the light emitting element **106**.

[0030] The light emitting element **106** may be at least disposed in the opening **104**O, and the light emitting element **106** includes a first electrode **112**, a light emitting layer **114** disposed on the first electrode **112**, and a second electrode **116** disposed on the light emitting layer **114**. Furthermore, the inside of the light emitting element **106** may include an electron injection layer (EIL), an electron transport layer (ETL), a charge generation layer (CGL), a hole injection layer (HIL), and a hole transport layer (HTL), and in order to clearly illustrate the light emitting element **106**, FIG. **1** omits these layers, but not limited thereto. In one embodiment, the first electrode **112** may be partially disposed in the connecting hole **110**C to be electrically connected to the drain (source) SD**2** of the driving element **110**D.

[0031] In the present embodiment, the first electrode **112** and the second electrode **116** may be an anode and a cathode respectively, but not limited thereto, and they may be exchanged. It is noted that since the part where the first electrode **112**, the light emitting layer **114** and the second electrode **116** are in contact may generate excitation light, the size of the light emitting element **106** may be defined as the area where the light emitting layer **114** contacts both the first electrode **112** and the second electrode **116**. The light emitting layer **114** and the second electrode **116** are sequentially at least disposed on the first electrode **112**, so a region of the light emitting element **106** that can generate excitation light may substantially correspond to the size of one first electrode **112**, but not limited thereto. If the light emitting layer **114** and the second electrode **116** extend or be widened to cover a plurality of the first electrodes **112**, the region where can generate excitation light still be the region of the light emitting layer **114** that both contacts the first electrodes **112** and the second electrode **116**, and the region for generating excitation light may substantially correspond to a sum of the areas of the plural first electrodes **112**. In some embodiments, the light emitting layer **114** and the second electrode **116** covering the plural first electrodes **112** may include discontinuous blocks respectively corresponding to the plural first electrodes **112**.

[0032] Because the first electrode **112** is formed after the first insulating layer **104** is formed, the first electrode **112** may include a side portion **112**S located on the side surface **104**S of the first insulating layer **104**, and a bottom portion **112**B not disposed on the side surface **104**S. In other words, the bottom portion **112**B of the first electrode **112** is disposed on the bottom of the recess R. In the present embodiment, the first electrode **112** extends along the side surface **104**S of the opening **104**O and the bottom of the recess R, and is a non-flat shape structure, such as a concave cup shape in a cross-sectional view, but is not limited thereto. For this reason, the light emitting element **106** formed by the first electrode **112**, and the light emitting layer **114** and the second electrode **116** covering the first electrode **112** may have an non-flat shape, for example a concave cup shape in a cross-sectional view.

[0033] It is noted that in the method of forming the conventional light emitting element, the first insulating layer **104** is firstly formed on the thin film transistor layer **110**, and then the first electrode **112** is formed. However, in the present embodiment, since the light emitting element **106** may have the non-flat shape, the light emitting element **106** may include larger light emitting area or reflective area as compared to the conventional light emitting element, but not limited thereto. Accordingly, the intensity of the excitation light generated by the light emitting element **106** may be increased or the utility of light after being reflected may be improved. Or, the converted light generated by the light conversion element **108** may be reflected.

[0034] In other embodiments, the side portion **112**S of the first electrode **112** may further extend to the top surface **104**T of the first insulating layer **104**, so that problems of the uneven edge, deckle edge, ragged edge, or suspended edge of the side portion, uneven thickness of the side portion, or inconsistency of the positions of the edges of different side portions resulted from processes (such as etching process) may be improved.

[0035] In some embodiments, heights H**1**, H**1'** of the side portions **112**S of the first electrode **112** disposed on the side surface **104**S of the first insulating layer **104** may be the same or different, for example the heights H**1**, H**1'** of the side portions **112**S on a side surface part **104S1** and another side surface part **104S2** may be the same or different. As the central line CL of the opening **104**O is regarded as a symmetry axis, a cross-sectional view of the first electrode **112** taken along the first direction D**1** may include a symmetric structure or asymmetric structure, but not limited thereto. The "height" comparison mentioned in this disclosure is relative to the same level, and drawings illustrate the heights are for example relative to the top surface **102**S of the substrate **102** that is flattened, wherein when the substrate **102** is the flexible substrate, the comparison mentioned above is relative to the top surface **102**S of the substrate **102** that is spread to be flat, but not limited thereto. In some embodiments, the cross-sectional view of the first electrode **112** taken along the second direction D**2** may include symmetric structure or asymmetric structure with respect to the central line CL, but not limited thereto. In some embodiments, a partial cross-sectional view of the first electrode **112** taken along the first direction D**1** and a partial cross-sectional view of the first electrode **112** taken along the second direction D**2** may be symmetric or asymmetric to each other, but not limited thereto. The first direction D**1** and the second direction D**2** may be different, for example be perpendicular to each other, but not limited thereto. In addition, the position of the central line CL may be obtained for example by obtaining a cross-sectional view of the display device **1** along a direction, then finding the bottom portion **112**B of the first electrode **112** from the cross-sectional view and taking the intermediate position between two ends of the bottom portion **112**B close to the first insulating layer **104** (e.g. boundary between the side surface part **104**S**1** of the first insulating layer **104** and the thin film transistor layer **110** and boundary between the side surface part **104**S**2** of the first insulating layer **104** and the thin film transistor layer **110**) as the central line CL, but not limited thereto. In some embodiments, the cross-sectional view of the display device **1** for finding the position of the central line CL may be taken along a cross-sectional line passing through at least a part of the driving element **110**D. In some embodiments, the cross-sectional direction of the cross-sectional view mentioned above may for example be the cross-section line L**3** along the first direction D**1** shown in FIG. **1**. The cross-sectional line L**3** may pass through both side surfaces **104**S of the first insulating layer **104**, or may pass through at least a part of the driving element **110**D (e.g. source (drain) SD**1**/drain (source) SD**2**), or pass through the connecting hole **110***c*, but not limited thereto.

[0036] In some embodiments, an angle θ**1** between the side portion **112**S and the bottom portion **112**B of the first electrode **112** may for example be greater than or equal to 90 degrees and less than or equal to 150 degrees ($90 \leq \theta 1 \leq 150$). The angle θ**1** may for example determined by adjusting an angle between the side surface **104**S of the first insulating layer **104** and the top surface **110**T of the thin film transistor layer **110**, but not limited thereto. In some embodiments, the angle θ**1** may be determined by the side portion **112**S and the bottom portion **112**B of the first electrode **112** or by the first insulating layer **104** and the thin film transistor layer **110**. When the top surface **104**T or the side surface **104**S of the first insulating layer **104** is an uneven surface, the calculating method of the angle θ**1** may refer to the description corresponding to FIG. **14**. Also, angles θ**1** between the bottom portion **112**B and different parts of the side portion **112**S of the first electrode **112** (e.g. parts of the side portion **112**S located on two sides of the bottom portion **112**B) may be different, and through adjusting the difference of the angles θ**1**, the coupling capacitance between the first electrode **112** and different elements in the underlying thin film transistor layer **110** (e.g. storage capacitance between the first electrode **112** and the common line (not shown) under the first electrode **112**), angle of reflecting the excitation light or the converted light, or the light emitting angle of the light emitting element **106** may be altered.

[0037] The first electrode **112** may include conductive material having reflective characteristics, such as silver (Ag), magnesium (Mg), aluminum (Al), platinum (Pt), palladium (Pd), gold (Au), nickel (Ni), neodymium (Nd), iridium (Ir), chromium (Cr), other suitable metal, or combinations of above metals. The second electrode **116** may include transparent or translucent conductive material, such as silver (Ag), aluminum (Al), ytterbium (Yb), titanium (Ti), magnesium (Mg), nickel (Ni), lithium (Li), calcium (Ca), copper (Cu), lithium fluorine/gallium (LiF/Ga), lithium fluorine/aluminum (LiF/Al), magnesium silver (MgAg), calcium silver (CaAg), nano silver glue, other suitable conductive material, or any combination of above conductive materials. Since the second electrode **116** has a thickness of for example several nanometers up to several ten nanometers, the second electrode **116** may allow light to penetrate.

[0038] In some embodiments, the light emitting layer **114** of the light emitting element **106** may include a single layer or multiple layers, and is not limited thereto. In some embodiments, the light emitting element **106** may further include the hole transport layer, the hole injection layer, the electron transport layer, the electron injection layer, and the charge generation layer covering the first electrode **112**, but not limited thereto. When the first electrode **112** is the anode, and the second electrode **116** is the cathode, the hole transport layer and the hole injection layer are disposed between the first electrode **112** and the light emitting layer **114**, and the electron transport layer and the electron injection layer are disposed between the second electrode **116** and the light emitting layer **114**. When the light emitting element **106** includes a plurality of light emitting layers **114**, the charge generation layer may be disposed between two light emitting layers **114**, but not limited thereto.

[0039] The light conversion element **108** is disposed on the light emitting element **106**. The light conversion element **108** may for example include phosphor material, fluorescent material, quantum dot particles or other light converting material that is capable of converting color of light, and the light converting materials mentioned above may be arbitrarily combined, but not limited thereto. Specifically, the light conversion element **108** may at least fill into the recess R, such that the height H**1** of the side portion **112**S of the first electrode **112** may be greater than the height H**2** of the bottom of the light conversion element **108**. Thus, both the side portion **112**S and the bottom portion **112**B of the first electrode **112** may reflect the converted light generated by the light conversion element **108**, thereby reducing leakage or loss of the converted light due to being emitted from the side surface **104**S of the first insulating layer **104** or improving light utility of the light conversion element **108**. Furthermore, by the design of the side portion **112**S of the first electrode **112**, the excitation light generated by the light emitting element **106** corresponding to the side portion **112**S may enter the light conversion element **108** from the side surface of the light conversion element **108**, thereby increasing light emitting area. Accordingly, the light conversion particles in the light conversion element **108** may be fully utilized, or the converting efficiency of the light conversion element **108** may be increased. For example, in order to have enough light conversion particles, the thickness of the light conversion element **108** may range from about 6 microns (μm) to 8 microns. When the height H**1** and the height H**1'** are different, a higher one of the height H**1** and the height H**1'** may be taken as the height H**1** to be compared with the height H**2**.

[0040] In some embodiments, the height H**3** of the top surface **108**T of the light conversion element **108** may be greater than the height H**1** of the side portion **112**S of the first electrode **112**, such that all the excitation light generated by the light emitting element **106** may be emitted into the light conversion element **108**. In this disclosure, the height H**1** of the side portion **112**S of the first electrode **112**, the height H**2** of the bottom of the light conversion element **108**, the height H**3** of the top surface **108**T of the light conversion element **108**, and the height H**4** of the top surface **104**T of the first insulating layer **104** may be defined as starting from the same level, for example starting from the top surface **102**S of the substrate **102**, but not limited thereto. When the top surface **108**T of the light conversion element **108** is arc-shaped, the height H**3** may be taken as the distance from bottom of the top surface **108**S of the light conversion element **108** closest to the substrate **102** to the top surface **102**S of the substrate **102**. In some embodiments, in order to avoid the light conversion element **108** flowing over the recess R during forming the light conversion element **108**, the height H**3** of the top surface **108**T of the light conversion element **108** may be less than the height H**4** of the top surface **104**T of the first insulating layer **104**.

[0041] The display device of the present disclosure is not limited to the above-mentioned embodiments and may include different embodiments or variant embodiments. In order to simplify the description, the elements of different embodiments and variant embodiments and the same element of the first embodiment will use the same label. In order to compare the difference between the first embodiment and different embodiments and variant embodiments, the following contents would focus on the difference between different embodiments or variant embodiments, and the repeated portion will not be redundantly described.

[0042] FIG. **2** schematically illustrates a cross-sectional view of a display device according to a second embodiment of the present disclosure. As shown in FIG. **2**, the difference between the display device **2** of the present embodiment and the first embodiment shown in FIG. **1** is that the display device **2** may further optionally include a protection layer **220** disposed between the light emitting element **206** and the light conversion element **208**. In detail, the protection layer **220** may include a stack of an inorganic material layer **220*a***, an organic material layer **220*b***, and an inorganic material layer **220*a*** to reduce moisture or oxygen from penetration. Besides, the protection layer **220** may also prevent the light emitting element **206** from being damaged by the filling process of the light conversion element **208**. For example, the inorganic material layer may include silicon nitride, silicon oxygen, silicon oxynitride, aluminum oxide, other suitable protecting material, or any combination of the inorganic materials mentioned above, but not limited thereto. The organic material layer may include resin, but not limited thereto. In some embodiments, the protection layer **220** may be single inorganic material layer **220*a*** or a stack of plural inorganic material layers. In some embodiments, another protection layer may be disposed on the light conversion element **208** to protect the light conversion element **208**.

[0043] In some embodiments, the side portion **212**S of the first electrode **212** may further extend to the top surface **104**T of the first insulating layer **104**, so that problems of the uneven edge, deckle edge, ragged edge, or suspended edge of the side portion, uneven thickness of the side portion, or inconsistency of the positions of the edges of different side portions resulted from processes (such as etching process) may be improved. Furthermore, the side portion **212**S may include a part **212**Sa extending to the top surface **104**T and another part **212**Sb not extending to the top surface **104**T. For example, an angle θ2 between the part **212**Sa and the part **212**Sb may be greater than or equal to 190 degrees and less than or equal to 275 degrees ($190 \leq \theta 2 \leq 275$), such that the first electrode **212** may be tightly adhered on the first insulating layer **104**, and peeling of the first electrode **212** from the first insulating layer **104** may be reduced. The angle θ2 may be located between a virtual line L**1** extending from the top surface of the part **212**Sb of the side portion **212**S and a virtual line L**2** extending from the top surface of the part **212**Sa of the side portion **212**S. When the top surface **104**T or the side surface **104**S of the first insulating layer **104** is uneven, the definitions of the top surface **104**T and the side surface **104**S and the calculating method of the angle θ2 may refer to the description corresponding to FIG. **14**.

[0044] In this embodiment, the display device **2** may further include a light blocking layer **222** disposed on the protection layer **220**, and the light blocking layer **222** has an opening **222**O corresponding to the opening **104**O of the first insulating layer **104**. The opening **222**O corresponding to the opening **104**O means that in the top view direction VD, the projection region of the bottom of the opening **222**O and the projection region of the bottom of the opening **104**O at least partially or completely overlap. In other words, the light blocking layer **222** is disposed on the top surface **104**T of the first insulating layer **104**. The light blocking layer **222** may shield light reflected by the part **212**Sa of the first electrode **212** extending to the top surface **104**T of the first insulating layer **104** and also shield the excitation light generated by the light emitting element **206** corresponding to the part **212**Sa, thereby avoiding mixing light. Furthermore, by means of the disposition of the light blocking layer **222**, the height H**3** of the top surface **208**T of the light conversion element **208** may be greater than the height H**4** of the top surface **104**T of the first insulating layer **104**, and the light conversion element **208** will not overflow into other openings **104**O. In some embodiments, the top surface **208**T of the light conversion element **208** may be convex, but not limited thereto. In a cross-sectional view of one embodiment, two sides of the top surface **208**T of the light conversion element **208** that contact the light blocking layer **222** may have different heights, and the height H**3** is the lower one of the different heights. In a cross-sectional view of one embodiment, the top surface **104**T of the first insulating layer **104** may have different heights, and the height H**4** is the higher one of the different heights.

[0045] FIG. **3** schematically illustrates a cross-sectional view of a display device according to a third embodiment of the present disclosure. As shown in FIG. **3**, the difference between the display device **3** of the present embodiment and the second embodiment shown in FIG. **2** is that the display device **3** may further optionally include a second insulating layer **324** disposed on the first insulating layer **104**, such that the part **212**Sa of the side portion **212**S of the first electrode **212** is disposed between the first insulating layer **104** and the second insulating layer **324**. Specifically, the second insulating layer **324** may be disposed between the first insulating layer **104** and the light emitting layer **114**, such that the part of the light emitting layer **114** located on the second insulating layer **324** may be separated from the part **212**Sa of the first electrode **212**, and the part of the light emitting layer **114** on the second insulating layer **324** will not generate light. Accordingly, the light emitting element **306** of the present embodiment may be formed of the part **212**Sb of the first electrode **212** contacting the light emitting layer **114**, the part of the light emitting layer **114** contacting the first electrode **212** and the part of the second electrode **116** contacting the part of the light emitting layer **114**. The angle between the part **212**Sa and the part **212**Sb (as the angle θ**2** shown in FIG. **2**) may be greater than or equal to 190 degrees and less than or equal to 275 degrees. In some embodiments, the second insulating layer **324** may include light blocking material for shielding the excitation light generated by the light emitting element **306** from laterally emitting out, thereby reducing light mixing. Accordingly, the display device **3** may not require the light blocking layer. In some embodiments, the second insulating layer **324** may be multilayer structure, so that the height of the top surface of the second insulating layer **324** may be increased to prevent the light conversion element **308** from overflowing in the situation that no light blocking layer is required. In some embodiments, the top surface **308**T of the light conversion element **308** may be concave, but not limited thereto. In some embodiments, the height H**1** of the part **212**Sa of the side portion **212**S of the first electrode **212** may be the same as the height H**1'** of the part **212**Sb of the side portion **212**S of the first electrode **212**. In such situation, the widths of the part **212**Sa and the part **212**Sb may optionally be the same, such that the first electrode **212** may include symmetric structure with respect to the central line CL, but not limited thereto. In a cross-sectional view of the embodiment shown in FIG. **3**, the height H**3** of the top surface **308**T of the light conversion element **308** may substantially be the height of the lowest point of the top surface **308**T in the cross-sectional view.

[0046] FIG. **4** schematically illustrates a cross-sectional view of a display device according to a fourth embodiment of the present disclosure. As shown in FIG. **4**, the difference between the display device **4** of the present embodiment and the first embodiment shown in FIG. **1** is that the thin film transistor layer **410** of the display device **4** may have a protrusion part **410***a* corresponding to the opening **104**O. In other words, the protrusion part **410***a* is disposed on an extending plane of the flat top surface **410**T of the thin film transistor layer **410** (as the virtual line at the bottom of the protrusion part **410***a* shown in FIG. **4**), and the top surface of the protrusion part **410***a* may form a part of the bottom of the recess R. For example, the protrusion part **410***a* may be formed for example by utilizing gray-tone mask or half-tone mask when patterning the planarization layer **410**F, but not limited thereto. By means of the protrusion part **410***a*, the area of the bottom portion **412**B of the first electrode **412** disposed in the recess R may be increased, thereby raising the intensity of the excitation light generated by the light emitting element **406**.

[0047] In addition, in some embodiments, the height of the side portion **412**S of the first electrode **412** disposed on the side surface **104**S of the first insulating layer **104** may not be identical, for example in the situation that the side portion **412**S doesn't extend to the top surface **104**T of the first insulating layer **104**, the height H**1** of the part **412S1** of the first electrode **412** (e.g. on the left side of the light conversion element **408**) may be less than the height H**1'** of the part **412S2** of the first electrode **412** (e.g. on the right side of the light conversion element **408**), so that the cross-sectional view of the first electrode **112** along the first direction D**1** is an asymmetric structure when the central line CL of the opening **104**O is used as the symmetry axis, but not limited thereto.

[0048] FIG. **5** schematically illustrates a cross-sectional view of a display device according to a fifth embodiment of the present disclosure. As shown in FIG. **5**, the difference between the display device **5** of the present embodiment and the fourth embodiment of FIG. **4** is that the thin film transistor layer **510** of the display device **5** may have a recess part **510***a* corresponding to the opening **104**O. The top surface **510**T of the recess part **510***a* may form a part of the bottom of the recess R. For example, the planarization layer **510**F of the thin film transistor layer **510** may have the recess part **510***a*. The recess part **510***a* may be formed for example by utilizing gray-tone mask or half-tone mask when patterning the planarization layer **510**F, but not limited thereto. By means of the recess part **510***a*, the area of the bottom portion **512**B of the first electrode **512** disposed on the top surface **510**T of the thin film transistor layer **510** may be increased, thereby raising the intensity of the excitation light generated by the light emitting element **506**.

[0049] Moreover, in some embodiments, a part of the side portion **512**S of the first electrode **512** disposed on the side surface **104**S of the first insulating layer **104** may extend to the top surface **104**T of the first insulating layer **104**. For example, the part **512S2** of the side portion **512**S located on the side surface part **104S2** (located on the right side of the light conversion element **108**) may extend to the top surface **104**T of the first insulating layer **104**, so that the part **512S2** may further include a part **512S2***a* located on the top surface **104**T of the first insulating layer **104** and a part **512S2***b* located on the side surface **104**S of the first insulating layer **104**. Also, the part **512S1** of the side portion **512**S located on the side surface part **104S1** (located on the left side of the light conversion element **108**) doesn't extend to the top surface **104**T of the first insulating layer **104**, so the height H**1** of the part **512S1** of the side portion **512**S may be less than the height H**1'** of the part **512S2** of the side portion **512**S on the top surface **104**T of the first insulating layer **104**. Accordingly, the cross-sectional view of the first electrode **512** along the first direction D**1** may be asymmetric with respect to the central line CL of the opening **104**O, but not limited thereto. In order to shield the excitation light generated by the light emitting element **506** disposed on the top surface **104**T of the first insulating layer **104**, a light blocking layer **522** may be further disposed on the top surface **104**T of the first insulating layer **104**. The light blocking layer **522** may have an opening **522**O corresponding to the opening **104**O of the first insulating layer **104**. Since the light blocking layer **522** of the present embodiment may be similar to or the same as the light blocking layer of the second embodiment, the light blocking layer **522** will not be redundantly detailed.

[0050] FIG. **6** schematically illustrates a cross-sectional view of a display device according to a sixth embodiment of the present disclosure. As shown in FIG. **6**, the difference between the display device **6** of the present embodiment and the embodiment shown in FIG. **4** is that the thin film transistor layer **610** of the display device **6** may have a plurality of protrusion parts **610***a* corresponding to the opening **104**O. In other words, the protrusion parts **610***a* are disposed on the extending plane of the flat top surface **610T1** of the thin film transistor **610** (as the virtual line located on the bottom of the protrusion parts **610***a* shown in FIG. **6**), and the top surfaces **610**aT of the protrusion parts **610**a may form a part of the recess R. Specifically, the planarization layer **610**F of the thin film transistor layer **610** may include a plurality of protrusion parts **610**a, and in the present embodiment, the plural protrusion parts **610**a are connected in sequence, such that the top surface **610**T**2** corresponding to the opening **104**O is formed to be wave-shaped by the top surfaces **610**aT of the plural protrusion parts **610**a. The protrusion parts **610**a may be formed for example by utilizing gray-tone mask or half-tone mask when patterning the planarization layer **610**F, but not limited thereto. Through the wave-shaped top surface **610**T**2**, the area of the bottom portion **612**B of the first electrode **612** disposed on the top surface **610**T**2** of the thin film transistor **610** may be increased, thereby increasing the intensity of the excitation light generated by the light emitting element **606**. In some embodiments, the thin film transistor layer **610** may include a plurality of recess parts (not shown) sequentially connected, such that the surfaces of the recess parts may be connected and form a wave-shaped top surface. When the thin film transistor layer **610** has the plural protrusion parts **610**a or plural recess parts connected in sequence, the height H**2** of the bottom of the light conversion element **108** may be taken for example as the lowest point of the light conversion element **108** between two adjacent protrusion parts **610**a or the lowest point of one of the recess parts.

[0051] In some embodiments, the part **612S1** of the side portion **612** of the first electrode **612** located on the side surface part **104S1** (located on the right side of the light conversion element **106**) may extend to the top surface **104**T of the first insulating layer **104**, so that the part **612S1** may further include a part **612S1**a disposed on the top surface **104**T of the first insulating layer **104** and a part **612S1**b disposed on the side surface **104S1** of the first insulating layer **104**. Also, the part **612S2** of the side portion **612S** located on the side surface part **104S2** (located on the right side of the light conversion element **106**) doesn't extend to the top surface **104**T of the first insulating layer **104**, and therefore, the height H**1** of the part **612S1** of the side portion **612S** may be greater than the height H**1'** of the part **612S2** of the side portion **612S**. In some embodiments, the display device **6** may further include a second insulating layer **624** disposed on the top surface **104**T of the first insulating layer **104**, and the part **612S1**a of the side portion **612S** extending to the top surface **104**T of the first insulating layer **104** is disposed between the first insulating layer **104** and the second insulating layer **624**. Since the second insulating layer **624** may be similar to or the same as the second insulating layer of the third embodiment, the second insulating layer **624** will not be redundantly detailed.

[0052] FIG. **7** schematically illustrates a cross-sectional view of a display device according to a seventh embodiment of the present disclosure. As shown in FIG. **7**, the difference between the display device **7** of the present embodiment and the second embodiment shown in FIG. **2** is that a first insulating layer **704** may further include a plurality of bumps **704**P**2**, wherein bumps **704**P**1** (that are similar to the first insulating layer **104** shown in FIG. **2**) of the first insulating layer **704** overlap the light blocking layer **722** in the top view direction VD and define pixels or sub-pixels of the display device **7**, and the bumps **704**P**2** doesn't overlap the light blocking layer **722** and are used for increasing the light emitting area of the light emitting element **706**. Furthermore, any adjacent two of the bumps **704**P**1** and bumps **704**P**2** may have an opening **704**O therebetween, wherein the opening **704**O is surrounded by the side surface **704S1**, the side surface **704S2** or/and the side surface of the first insulating layer **704**. The opening **722**O of the light blocking layer **722** may correspond to the plural openings **704**O of the first insulating layer **704**, and the first electrode **704** may extend into at least two of the openings **704**O, but not limited thereto. In other words, the first electrode **712** may cross at least one bump **704**P**2**. More specifically, the first electrode **712** may extend from the side surface **704S2** of the bump **704**P**2** (i.e. from one of the openings **704**O) to another side surface **704S1** (i.e. into another opening **704**O) through the top surface **704**T of the bump **704**P**2**, thereby increasing the area of the first electrode **712**. When the light emitting layer **114** and the second electrode **116** are continuous layers, the light emitting element **706** may cross the plural openings **704**O, and therefore, the light emitting area of the light emitting element **706** may be increased to improving the intensity of the excitation light generated by the light emitting element **706**.

[0053] In the present embodiment, the light conversion element **708** may be disposed in the openings **704**O of the first insulating layer **704** and at least partially be disposed in the opening **722**O of the light blocking layer **722**, so as to be disposed on the light emitting element **706**. In the present embodiment, one opening **722**O of the light blocking layer **722** may correspond to three openings **704**O of the first insulating layer **704**, and the first electrode **712** may cross three openings **704**O and/or cross two bumps **704**P**2**, but not limited thereto. The openings **704**O of the first insulating layer **704**, the corresponding opening **722**O of the light blocking layer **722**, and the top surface **110**T of the thin film transistor layer **110** may form the recess R. In the present embodiment, the height H**4'** of the bump **704**P**2** may be the same as the height H**4** of the bump **704**P**1** located under the light blocking layer **722**, such that the height of the part of the first electrode **712** on the bump **704**P**2** may be greater than the height H**1** of the side portion **712S** of the first electrode **712**, but not limited thereto. In some embodiments, the height H**4'** of the bump **704**P**2** may be less than the height H**4** of the bump **704**P**1** under the light blocking layer **722**.

[0054] FIG. **8** schematically illustrates a cross-sectional view of a display device according to an eighth embodiment of the present disclosure. As compared to the first embodiment of FIG. **1**, FIG. **8** further exemplifies layers on the light conversion elements and regions of the light conversion elements corresponding to different colors. As shown in FIG. **8**, the display device **8** of the present embodiment may include at least three light emitting elements (e.g. a light emitting element **1061**, a light emitting element **1062**, and a light emitting element **1063**), the first insulating layer **104** may include openings (e.g. an opening **104O1**, an opening **104O2**, and an opening **104O3**), and the light emitting element **1061**, the light emitting element **1062**, and the light emitting element **1063** may be at least disposed in the opening **104O1**, opening **104O2**, and opening **104O3** respectively. In the present embodiment, the light emitting element **1061**, the light emitting element **1062** and the light emitting element **1063** may respectively include a first electrode **112**, and the first electrodes **112** correspond to the opening **104O1**, opening **104O2**, and opening **104O3** respectively. The height H**11**, the height H**12**, and the height H**13** of the side portions **112S** of the first electrodes **112** of the light emitting element **1061**, the light emitting element **1062**, and the light emitting element **1063** may be the same, but not limited thereto. In some embodiments, the height H**11**, the height H**12**, and the height H**13** of the side portions **112**S of different first electrodes **112** may be different or at least two of which are different. In some embodiments, the side portion **112**S of at least one of the first electrodes **112** may adopt the side portion of the first electrode of any one embodiment mentioned above.

[0055] The light emitting element **1061**, the light emitting element **1062**, and the light emitting element **1063** of the present embodiment may share the same light emitting layer **114** and the same second electrode **116**. In other words, the light emitting layer **114** and the second electrode **116** may be continuous layers and cover the first insulating layer **104** and the first electrode **112**, such that the light emitting element **1061**, the light emitting element **1062**, and the light emitting element **1063** may generate the excitation light with the same color, such as blue light, but the present disclosure is not limited thereto. In some embodiments, the light emitting layer **114** may be a discontinuous layer, such that portions of the light emitting layer **114** corresponding to different light emitting elements may be used to generate different colors respectively. The color generated by the light emitting element **1063** regarded as a color of a sub-pixel is taken for an example in the present embodiment, and the display device **7** may include the light conversion element **1081** and the light conversion element **1082** for generating converted light of different colors, and a light scattering layer **826** disposed on the light emitting element **1063**. The light conversion element **1081** and the light conversion element **1082** are disposed on the light emitting element **1061** and the light emitting element **1062** respectively. The light conversion element **1061** and the light conversion element **1062** may convert the excitation light generated by the light emitting element **1061** and the light emitting element **1062** into the converted light of corresponding colors, for example generate the converted light of red and green respectively.

[0056] The light scattering layer **826** may for example include a plurality of scattering particles, but not limited thereto. The material of the scattering particle may for example include titanium oxide or a structural particle with scattering characteristics, but not limited thereto. In some embodiments, when the color of the light emitting element **1063** is not used as the color of the sub-pixel (e.g. color of the sub-pixel of image viewed by user), each light emitting element (all the light emitting element **1061**, the light emitting element **1062**, and the light emitting element **1063**) has the light conversion element disposed thereon, and the light conversion elements corresponding to different light emitting elements (the light emitting element **1061**, the light emitting element **1062**, and the light emitting element **1063**) may generate the converted light of different colors. In such situation, the light conversion element may for example generate red light, green light, and blue light, and the light emitting element **1061**, the light emitting element **1062**, and the light emitting element **1063** may for example generate the excitation light with the wavelength less than the red light, the green light, and the blue light, such as blue light or ultraviolet light, but not limited thereto. In the present embodiment, the light emitting layer **114** may include a plurality of sub-layers **114***a*, for example the light emitting layer **114** of the present embodiment may include four sub-layers **114***a*, but not limited thereto. The sub-layers **114***a* for example include organic light emitting layer, hole transport layer, hole injection layer, electron transport layer, electron injection layer or charge generation layer, but not limited thereto. In some embodiments, the light emitting layer **114** may be a single-layer organic light emitting layer, but not limited thereto.

[0057] In the present embodiment, the display device **8** may further include a light blocking layer **822** disposed on the first insulating layer **104**, and the light blocking layer **822** may include openings (e.g. an opening **822O1**, an opening **822O2**, and an opening **822O3**) respectively corresponding to the opening **104O1**, the opening **104O2**, and the opening **104O3**, wherein each of the opening **822O1**, the opening **822O2**, and the opening **822O3** and the corresponding one of the opening **104O1**, the opening **104O2**, and the opening **104O3** form a part of sidewall of the corresponding one of the recess R**1**, the recess R**2**, and recess R**3**. The light blocking layer **822** may be used as a partition wall for preventing the light conversion element **1081**, the light conversion element **1082**, and the light scattering layer **826** from overflow. The light blocking layer **822** may be similar to or the same as the light blocking layer of the second embodiment and will not be repeatedly detailed.

[0058] The display device **8** of the present embodiment may further optionally include a protection layer **220** disposed between the light emitting element **1061** and the light conversion element **1081**, between the light emitting element **1062** and the light conversion element **1082** and between the light emitting element **1063** and the light scattering layer **826** and used for preventing the light emitting element **1061**, the light emitting element **1062** and the light emitting element **1063** from damage. The protection layer **220** of the present embodiment may be similar to or the same as the protection layer of the second embodiment and will not be redundantly mentioned.

[0059] The display device **8** may further optionally include a light filtering layer **828** that has the characteristics of selecting light with specific wavelength and is disposed on the light conversion element **1081** and the light conversion element **1082**, such that the excitation light with the specific wavelength generated by the light emitting element **1061** and the light emitting element **1062** may be blocked, inhibited, reflected, or recycled by the light filtering layer **828**, and the excitation light with the specific wavelength is prevented from being emitted out from the opening **822O1** and the opening **822O2**. Accordingly, purity of final output light emitted by the regions corresponding to the light conversion element **1081** and the light conversion element **1082** may be increased (i.e. the color of the final output light may be close to the color of the converted light generated by the corresponding one of the light conversion element **1081** and the light conversion element **1082**). The structure of the light filtering layer **828** may be multilayer and be a distributed Bragg reflector (DBR) formed of layers with different refractive indices alternately stacked or a color filter layer (such as a red photoresist, a yellow photoresist, a green photoresist, etc.), but not limited thereto. In the present embodiment, since the color generated by the light emitting element **1063** may be regarded as the color of one sub-pixel (e.g. the color of the sub-pixel of the image viewed by the user), no light filtering layer **828** may be disposed in the opening **822O3** of the light blocking layer **822**. In some embodiments, when the color of the light emitting element **1063** is not regarded as the color of the sub-pixel, the light filtering layer **828** may be disposed in the opening **822**O**3** of the light blocking layer **822**. In some embodiments, the light filtering layer **828** may further extend onto a part or all of the light blocking layer **822**.

[0060] In some embodiments, spaces of the opening **822**O**1**, the opening **822**O**2**, and the opening **822**O**3** that are not fully filled may further be filled with an encapsulation layer **830**, and the encapsulation layer **830** may further cover the light blocking layer **822**. The encapsulation layer **830** may for example include organic or inorganic encapsulating material which can block moisture and oxygen and/or have the characteristics of planarization layer for facilitating the following processes. Also, the encapsulation layer **830** may be an anti-reflection layer for anti-reflecting light from the outside environment. The display device **8** may further optionally include a cover layer **832** disposed on the encapsulation layer **830** and used for protecting the underlying layers.

[0061] In some embodiments, the display device **8** may further optionally include an auxiliary electrode **834** for reducing resistance difference between the second electrodes **116** of the light emitting element **1061**, the light emitting element **1062**, and the light emitting element **1063** and outer power source or peripheral circuit. For example, the auxiliary electrode **834** may be disposed between the second electrodes **116** and the protection layer **220** and located under the light blocking layer **822** and overlapped with the light blocking layer **822**. The material of the auxiliary electrode **834** may for example include MgAg layer, nano silver glue or other suitable conductive material. In some embodiments, the auxiliary electrode **834** and the second electrodes **116** may include the same material, but not limited thereto. In some embodiments, the auxiliary electrode **834** may be disposed on the light blocking layer **822**.

[0062] FIG. **9** schematically illustrates a cross-sectional view of a display device according to a ninth embodiment of the present disclosure. As shown in FIG. **9**, the difference between the display device **9** of the present embodiment and the eighth embodiment shown in FIG. **8** is that the height H**11** and the height H**12** of the side portions **912**S of the first electrodes **912** of different light emitting element **9061** and light emitting element **9062** may not be the same. Specifically, the side portion **912**S of the first electrode **912** of the light emitting element **9062** may extend to the top surface **104**T of the first insulating layer **104**, and therefore, may include a part **912**Sa located on the top surface **104**T of the first insulating layer **104** and another part **912**Sb located on the side surface **104**S of the first insulating layer **104**. Also, the side portion **912**S of the first electrode **912** of the light emitting element **9061** doesn't extend to the top surface **104**T of the first insulating layer **104**, so the height H**12** of the side portion **912**S of the first electrode **912** of the light emitting element **9062** may be greater than the height H**11** of the side portion **912**S of the first electrode **912** of the light emitting element **9061**. For this reason, as the top view direction VD is taken as the symmetry axis, the first electrode **912** of the light emitting element **9061** and the first electrode **912** of the light emitting element **9062** are not symmetric to each other. In other words, through altering the sizes of the first electrodes **912** of different light emitting element **9061** and light emitting element **9062** may change the light emitting brightness of the light emitting element **9061** and the light emitting element **9062**, thereby adjusting the white hue of the display device **9**. In the present embodiment, the first electrode **912** of the light emitting element **9063** may have the same structure as the first electrode **912** of the light emitting element **9061**, and therefore, the height H**11** of the side portion **912**S of the first electrode **912** of the light emitting element **9061** and the height H**13** of the side portion **912**S of the first electrode **912** of the light emitting element **9063** may be the same, but not limited thereto. In some embodiments, the height H**11** of the side portion **912**S of the first electrode **912** of the light emitting element **9061** and the height H**13** of the side portion **912**S of the first electrode **912** of the light emitting element **9063** may be different. In some embodiments, at least two of the first electrodes **912** of the light emitting element **9061**, the light emitting element **9062**, and the light emitting element **9063** may use the structures of the first electrodes **912** of the light emitting element **9061** and the light emitting element **9062**. In some embodiments, at least a part of each of the side portions **912**S of the first electrodes **912** of the light emitting element **9061**, the light emitting element **9062**, and the light emitting element **9063** may extend to the top surface **104**T of the first insulating layer **104**, and the first electrodes **912** of the light emitting element **9061**, the light emitting element **9062**, and the light emitting element **9063** may be electrically isolated.

[0063] In some embodiments, the light emitting layer **914** may be a discontinuous layer and include blocks (e.g. a block B**1**, a block B**2**, a block B**3**) separated from each other. In other words, the block B**1** of the light emitting layer **914** of the light emitting element **9061**, the block B**2** of the light emitting layer **914** of the light emitting element **9062**, and the block B**3** of the light emitting layer **914** of the light emitting element **9063** may be disconnected from each other. In such situation, the block B**1** of the light emitting layer **914** of the light emitting element **9061**, the block B**2** of the light emitting layer **914** of the light emitting element **9062**, and the block B**3** of the light emitting layer **914** of the light emitting element **9063** may for example be used for generating excitation light of different colors. In order to clearly illustrate the light emitting layer **914**, the light emitting layer **914** of FIG. **9** take single layer as an example, but not limited thereto. In some embodiments, the light emitting layer **914** may be multilayer structure.

[0064] FIG. **10** schematically illustrates a top view of a display device according to a tenth embodiment of the present disclosure, and FIG. **11** schematically illustrates a cross-sectional view of FIG. **10** taken along a cross-sectional line A-A'. As shown in FIG. **10** and FIG. **11**, the difference between the display device **10** of the present embodiment and the eighth embodiment shown in FIG. **8** is that one light conversion element **10081** or one light scattering layer **1026** of the display device **10** may be disposed on plural light emitting elements **106**. For example, the display device **10** may include a light conversion element **10081** and a light conversion element **10082** for generating converted light of different colors, and a light scattering layer **1026**. The following description further takes the light conversion element **10081** corresponding to three light emitting elements (e.g. the light emitting element **106-1**, the light emitting element **106-2**, and the light emitting element **106-3**) as an example, but not limited thereto. The light conversion element **10081** or the light scattering layer **1026** may correspond to plural light emitting elements. In the present embodiment, one opening **1022**O of the light blocking layer **1022** may correspond to three openings of the first insulating layer **104** (e.g. the opening **104**O-**1**, the opening **104**O-**2**, and the opening **104**O-**3**), the light emitting element **106**-**1**, the light emitting element **106**-**2**, and the light emitting element **106**-**3** are disposed corresponding to the opening **104**O-**1**, the opening **104**O-**2**, and the opening **104**O-**3** respectively, and the light conversion element **10081** fills up the opening **104**O-**1**, the opening **104**O-**2** and the opening **104**O-**3**. Accordingly, the filling of the light conversion element **10081** may be completed by performing a one-time injecting step or dripping step of inkjet process, thus reducing inkjet time.

[0065] In the present embodiment, each of the light emitting element **106**-**1**, the light emitting element **106**-**2**, and the light emitting element **106**-**3** is electrically connected to a corresponding driving element **110**D respectively. Since the light conversion element **10081** is disposed in the same opening **1022**O of the light blocking layer **1022**, the light emitting element **106**-**1**, the light emitting element **106**-**2**, and the light emitting element **106**-**3** located in the same opening **1022**O may correspond to the sub-pixels of the same color. The light conversion element **10081** may be divided into three portions (e.g. a portion **10081**-**1**, a portion **10081**-**2**, and a portion **10081**-**3**) located on the light emitting element **106**-**1**, the light emitting element **106**-**2**, and the light emitting element **106**-**3** respectively, and therefore, the light emitting element **106**-**1**, the light emitting element **106**-**2**, and the light emitting element **106**-**3** may correspond to the sub-pixels of the same color. For example, the portion **10081**-**1**, the portion **10081**-**2**, and the portion **10081**-**3** of the light conversion element **10081** may be divided by the opening **104**O-**1**, the opening **104**O-**2**, and the opening **104**O-**3** of the first insulating layer **104**. The virtual line located between any two of the portion **10081**-**1**, the portion **10081**-**2**, and the portion **10081**-**3** shown in FIG. **10** and FIG. **11** are for illustration but not intended to limit the boundary between any two of the portion **10081**-**1**, the portion **10081**-**2**, and the portion **10081**-**3**. Similarly, the light conversion element **10082** may for example be divided into three portions (e.g. a portion **10082**-**1**, a portion **10082**-**2**, and a portion **10082**-**3**) by the first insulating layer **104**, and the portion **10082**-**1**, the portion **10082**-**2**, and the portion **10082**-**3** correspond to the sub-pixels of the same color. The light scattering layer **1026** for example be divided into three portions (e.g. a portion **1026**-**1**, a portion **1026**-**2**, and a portion **1026**-**3**) by the first insulating layer **104**, and the portion **1026**-**1**, the portion **1026**-**2**, and the portion **1026**-**3** correspond to the sub-pixels of the same color. Through disposing the opening **104**O-**1**, the opening **104**O-**2**, and the opening **104**O-**3** to be adjacent, the resolution of the display device **10** may be effectively increased.

[0066] It is noted that during the inkjet process, a nozzle **1036***a*, a nozzle **1036***b*, and a nozzle **1036***c* corresponding to different colors may be moved along a direction D2 different from the horizontal direction HD, so a distance between the light conversion element **10081** and the light conversion element **10082** of different colors in the horizontal direction HD, and distances between the light scattering layer **1026** and the light conversion element **10081** and between the light scattering layer **1026** and the light conversion element **10082** in the horizontal direction HD may be reduced. Accordingly, distances between different sub-pixels in the horizontal direction HD may be shrunk, and the resolution of the display device **10** is increased.

[0067] FIG. **12** schematically illustrates a top view of a display device according to an eleventh embodiment of the present disclosure, and FIG. **13** schematically illustrates a cross-sectional view of FIG. **12** taken along a cross-sectional line B-B'. As shown in FIG. **12** and FIG. **13**, the difference between the display device **11** of the present embodiment and the tenth embodiment shown in FIG. **11** is that plural light emitting elements (e.g. the light emitting element **106'**, the light emitting element **106"**, and the light emitting element **106'"**) of the display device **11** corresponding to the same light conversion element **11081** or the light scattering layer **1126** may be electrically connected to the same driving element **1110**D. The light conversion element **11081** or the light scattering layer **1126** corresponding to three light emitting elements (the light emitting element **106'**, the light emitting element **106"**, and the light emitting element **106'"**) is taken for an example in the following description, but not limited thereto, and the light conversion element **11081** or the light scattering layer **1126** may correspond to plural light emitting elements **106**. Specifically, the portion **11081'**, the portion **11081"**, and the portion **11081'"** of the light conversion element **11081** may be disposed on the light emitting element **106'**, the light emitting element **106"**, and the light emitting element **106'"** respectively, the thin film transistor layer **1110** may include a plurality of connecting holes **1110**C, and through the connecting holes **1110**C, the light emitting element **106'**, the light emitting element **106"**, and the light emitting element **106'"** may be electrically connected to the electrode **1110**E (source or drain) of the same driving element **1110**D, such that the light emitting element **106'**, the light emitting element **106"**, and the light emitting element **106'"** may correspond to the same sub-pixel. Because the light emitting element **106'**, the light emitting element **106"**, and the light emitting element **106'"** are electrically connected to the same driving element **1110**D, the light emitting element **106'**, the light emitting element **106"**, and the light emitting element **106'"** can be driven by small current, thereby improving screen burn-in problem and increasing display quality and/or using lifetime. Similarly, at least one of the light conversion element **11082** and the light scattering layer **1126** may be disposed on the light emitting element **106'**, the light emitting element **106"**, and the light emitting element **106'"** that are electrically connected to the same driving element **110**D. For example, the three portions (the portion **11081'**, the portion **11081"**, and the portion **11081'"**) of the light conversion element **11081** may be divided by three openings (the opening **104'**, the opening **104"**, and the opening **104'"**) of the first insulating layer **104**. The virtual line located between any two of the portion **11081'**, the portion **11081"**, and the portion **11081'"** shown in FIG. **12** and FIG. **13** are for illustration but not intended to limit the boundary between any two of the portion **11081'**, the portion **11081"**, and the portion **11081'"**.

[0068] FIG. **14** schematically illustrates a cross-sectional view of a display device according to a twelfth embodiment of the present disclosure. As shown in FIG. **14**, the first insulating layer **1204** of the display device **12** of the present embodiment has an arc-shaped surface. It is noted that for convenience, the "height" comparison of the present embodiment is relative to the top surface **110**T of the thin film transistor layer **110** as an example, wherein when the substrate **102** is a flexible substrate, the comparison mentioned above is relative to the top surface **110**T of the substrate **102** after the substrate **102** is spread to be flat, but not limited thereto. When the first insulating layer **1204** has an uneven surface (such as arc-shaped), the height H**4** may for example be the height H**4** of the top point TP of the first insulating layer **1204** or an average of plural points that are arbitrary taken from the top surface **1204**T of the first insulating layer **1204** (e.g. taking 3 points), wherein the positions of the points may be taken without considering the part of the top surface **1204**T close to the side surface **1204**S. In some embodiments, when the first electrode **1212** is formed on the arc-shaped top surface **1204**T of the first insulating layer **1204** to have an arc-shaped surface, the height H**1** of the first electrode **1212** may be obtained by the method similar to the height H**4**. Also, when the first insulating layer **1204** has the arc-shaped surface, the top surface **1204**T of the first insulating layer **1204** may be defined as the surface between the height H**4** and the height H**4** of 0.95 times, and the surface that is located under the height H**4** of 0.95 times is defined as the side surface **1204**S. The angle between the top surface **1204**T and the side surface **1204**S of the first insulating layer **1204** may be obtained according to the following described method. Firstly, a reference point P**1** located at the bottom of the surface of the first insulating layer **1204** and a reference point P**2** at which the surface of the first insulating layer **1204** is located at the height of 0.95 times are taken, and then, an intermediate point SP between the reference point P**1** and the reference P**2** and on the surface of the first insulating layer **1204** is obtained. Next, an angle θ**3** between the top surface **1204**T and the side surface **1204**S of the first insulating layer **1204** may be defined as an angle between a tangent line L**4** at the top point TP and a tangent line L**5** at the intermediate point SP of the first insulating layer **1204**. When the first electrode **1212** is formed on the arc-shaped top surface **1204**T of the first insulating layer to have the arc-shaped surface, the top surface of the first electrode **1212** may be defined as the surface between the height H**1** and the height H**1** of 0.95 times, and the surface that is located under the height H**1** of 0.95 times is defined as the side surface. An angle θ**2** between a part **1212**Sa of the side portion **1212**S of the first electrode **1212** that extends to the top surface **1204**T and another part **1212**Sb of the side portion **1212**S of the first electrode **1212** that doesn't extend to the top surface **1204**T (as shown in FIG. **2**) may be calculated by the angle θ**3** of the first insulating layer **1204** or directly by the angle between the virtual line of the part **1212**Sa (as the virtual line L**2** shown in FIG. **2**) and the virtual line of the part **1212**Sb (as the virtual line L**1** shown in FIG. **2**) of the first electrode **1212**. Furthermore, in the present disclosure, when the side portion **1212**S of the first electrode **1212** extends to the top surface **1204**T of the first insulating layer **1204**, the top of the side portion **1212**S of the first electrode **1212** may be defined as a point of the top surface of the side portion **1212**S of the first electrode **1212** farthest from the layer on which the first electrode **1212** is disposed (such as the top surface **110**T of the thin film transistor layer **110**). When the side portion **1212**S of the first electrode **1212** is just disposed on the side surface **1204**S of the first insulating layer **1204** but doesn't extend to the top surface **1204**T, the top of the side portion **1212**S of the first electrode **1212** may be defined as a side of the side portion **1212**S of the first electrode **1212** farthest from the layer on which the first electrode **1212** is disposed (such as the top surface **110**T of the thin film transistor layer **110**).

[0069] For example, the angle between the part **1212**Sa and the part **1212**Sb of the first electrode **1212** may refer to the method of the angle between the top surface **1204**T and the side surface **1204**S of the first insulating layer **1204**, i.e. a reference point located at the bottom portion **1212**B of the surface of the first electrode **1212** and a reference point at which the surface of the first electrode **1212** is located at the height of 0.95 times are taken, an intermediate point between two reference points and on the surface of the first electrode **1212** is then obtained, and the angle θ**2** between the part **1212**Sa and the part **1212**Sb of the first electrode **1212** can be defined as the angle between a tangent line at the top point and a tangent line at the intermediate point of the first electrode **1212**. The part **1212**Sa of the first electrode **1212** may be defined as the part between the height H**1** and the height H**1** of 0.95 times, and the part located under the height H**1** of 0.95 times is defined as the part **1212**Sb.

[0070] Furthermore, since a part of the bottom of the light conversion element **108** close to the connecting hole **110**C may sink, when the cross-sectional line crosses the connecting hole **110**C, the height H**2** of the bottom of the light conversion element **108** may not consider the height of the part close to the connecting hole **110**C, and the height H**2** may just consider the height of the relatively flat region far away from the connecting hole **110**C.

[0071] Moreover, the side surface **1204**S of the first insulating layer **1204** may be an uneven surface. In the present embodiment, the first electrode **1212** formed on the arc-shaped side surface **1204**S of the first insulating layer **1204** may have arc-shaped side portion **1212**S. In such situation, the finding method of the angle θ**1** between the side portion **1212**S and the bottom portion **1212**B of the first electrode **1212** may be mentioned in the following description. Firstly, a reference point P**3** at which the side surface **1204**S of the first insulating layer **1204** is located at the height H**1** of 0.5 times is obtained. Then, the angle θ**1** between the side portion **1212**S and the bottom portion **1212**B of the first electrode **1212** may be defined as the angle between a tangent line L**6** of the side surface **1204**S of the first insulating layer **1204** at the reference P**3** and the top surface **110**T of the thin film transistor layer **110**.

[0072] In some embodiments, the technical features in the embodiments mentioned above may be arbitrarily mixed, combined or used when this combination doesn't depart from the spirit of the present disclosure or the technical features don't conflict to each other.

[0073] The display device provided by the present disclosure has the uneven first electrode, so the light emitting element may be uneven, thereby increasing the light emitting area. Accordingly, the excitation light generated by the light emitting element corresponding to the side portion of the first electrode may emit into the side surface of the light conversion element, such that the light conversion particles in the light conversion element may be fully used, thereby increasing the converting efficiency of the light conversion element. The uneven first electrode may reduce loss and leakage of the converted light generated by the light conversion element due to being emitted out from the side surface of the first insulating layer or may increase the utility of the converted light of the light conversion element, so the luminous efficiency of the display device can be improved.

[0074] Those skilled in the art will readily observe that numerous modifications and alterations of the device and method may be made while retaining the teachings of the disclosure. Accordingly, the above disclosure should be construed as limited only by the metes and bounds of the appended claims.

What is claimed is:

1. A display device, comprising:

a substrate;

an insulating layer disposed on the substrate and comprising at least one opening;

a light blocking layer disposed on the insulating layer;

a light emitting element, wherein at least a part of the light emitting element is disposed in the at least one opening;

a light conversion element disposed on the light emitting element;

a color filter layer disposed on the light conversion element; and

a protection layer disposed between the light conversion element and the light emitting element,

wherein in a cross-sectional view, the light blocking layer has a first maximum thickness, and the insulating layer has a second maximum thickness,

wherein the first maximum thickness is greater than the second maximum thickness,

wherein in the cross-sectional view, the light conversion element has a third maximum thickness, and the color filter layer has a fourth maximum thickness,

wherein the third maximum thickness is greater than the fourth maximum thickness.

2. The display device as claimed in claim **1**, wherein the protection layer comprises a plurality of inorganic material sub layers.

3. The display device as claimed in claim **1**, further comprising a cover layer, wherein the color filter layer is disposed between the light conversion element and the cover layer.

4. The display device as claimed in claim **1**, further comprising an anti-reflection layer disposed on the light conversion element.

5. The display device as claimed in claim **1**, further comprising a transistor electrically connected to the light emitting element, wherein the transistor overlaps the at least one opening.

6. The display device as claimed in claim **5**, further comprising a planarization layer disposed between the transistor and the insulating layer.

7. The display device as claimed in claim **1**, further comprising an auxiliary electrode disposed on the substrate and overlapped with the light blocking layer.

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